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667 (continued): ON THE BICIRCULAR QUARTIC.

the generating circle. If we consider the two points on the generating circle, and write dS' for the element at the other point, then we have

$$dS' = \frac{2\rho}{\sqrt{n}} d\omega,$$

which is Casey's formula $dS' - dS = 2\rho d\phi$ (273).

17. The foregoing forms of dX , dY are those which give most directly the required value of dS ; but I had previously obtained them in a different form. Writing

$$\Lambda = \beta x - \alpha y + (f - g)xy,$$

then

$$x\Lambda = \beta x^2 - \alpha xy + (f - g)x^2y,$$

or since $(f - g)x^2 = 1 - (g + \theta)r^2$, this is

$$x\Lambda = \beta x^2 - \alpha xy + \{1 - (g + \theta)(x^2 + y^2)\}y = y\{1 - \alpha x - \beta y\} + (x^2 + y^2)\{\beta - (g + \theta)y\},$$

that is, $= (x^2 + y^2)\{yR' + \beta - (g + \theta)y\} + y\sqrt{\Omega}$,

and similarly

$$\begin{aligned} x\Lambda - y\sqrt{\Omega} &= (x^2 + y^2)\{yR' + \beta - (g + \theta)y\}, \\ -y\Lambda - x\sqrt{\Omega} &= (x^2 + y^2)\{xR' + \alpha - (f + \theta)x\}. \end{aligned}$$

We have therefore

$$\begin{aligned} \frac{dX}{d\omega} &= \frac{R' d\omega}{(x^2 + y^2)\sqrt{\Omega}} (x\Lambda - y\sqrt{\Omega}), \\ \frac{dY}{d\omega} &= \frac{R' d\omega}{(x^2 + y^2)\sqrt{\Omega}} (y\Lambda + x\sqrt{\Omega}), \end{aligned}$$

and thence a value of dS which, compared with the former value, gives

$$\Omega + \Lambda^2 = (x^2 + y^2)h^2,$$

an equation which may be verified directly.

Formulae for the Inscribed Quadrilateral. Art. Nos. 18 to 22.

18. We consider on the curve four points, A, B, C, D , forming a quadrilateral, $ABCD$. The coordinates are taken to be (X, Y) , (X_1, Y_1) , (X_2, Y_2) , (X_3, Y_3) respectively. It is assumed that (A, B) , (B, C) , (C, D) , (D, A) belong to the generations 1, 2, 3, 0, and depend on the parameters (x_1, y_1) , (x_2, y_2) , (x_3, y_3) , (x, y) respectively.

We write

$$\begin{aligned}\Omega &= (1 - \alpha x - \beta y)^2 - \gamma^2(x^2 + y^2), \\ \Omega_1 &= (1 - \alpha_1 x_1 - \beta_1 y_1)^2 - \gamma_1^2(x_1^2 + y_1^2), \\ \Omega_2 &= (1 - \alpha_2 x_2 - \beta_2 y_2)^2 - \gamma_2^2(x_2^2 + y_2^2), \\ \Omega_3 &= (1 - \alpha_3 x_3 - \beta_3 y_3)^2 - \gamma_3^2(x_3^2 + y_3^2),\end{aligned}$$

and then, $\sqrt{\Omega}$ denoting as above a determinate value, positive or negative as the case may be, of the radical, and similarly $\sqrt{\Omega_1}, \sqrt{\Omega_2}, \sqrt{\Omega_3}$ denoting determinate values of these radicals respectively, each radical having its own sign at pleasure, we further write

$$\begin{aligned} (x^2 + y^2) R' &= 1 - \alpha x - \beta y + \sqrt{\Omega}, & (x_1^2 + y_1^2) R'_1 &= 1 - \alpha_1 x_1 - \beta_1 y_1 + \sqrt{\Omega_1}, \\ (x_1^2 + y_1^2) R_1 &= 1 - \alpha_1 x_1 - \beta_1 y_1 - \sqrt{\Omega_1}, & (x_2^2 + y_2^2) R'_2 &= 1 - \alpha_2 x_2 - \beta_2 y_2 + \sqrt{\Omega_2}, \\ (x_2^2 + y_2^2) R_2 &= 1 - \alpha_2 x_2 - \beta_2 y_2 - \sqrt{\Omega_2}, & (x_3^2 + y_3^2) R'_3 &= 1 - \alpha_3 x_3 - \beta_3 y_3 + \sqrt{\Omega_3}, \\ (x_3^2 + y_3^2) R_3 &= 1 - \alpha_3 x_3 - \beta_3 y_3 - \sqrt{\Omega_3}, & (x^2 + y^2) R &= 1 - \alpha x - \beta y - \sqrt{\Omega}; \end{aligned}$$

and this being so, we must have

$$\begin{aligned} X &= \alpha + R'x = \alpha_1 + R_1x_1, & Y &= \beta + R'y = \beta_1 + R_1y_1, & R' &= \frac{1}{2}(X^2 + Y^2 - k), & R_1 &= \frac{1}{2}(X^2 + Y^2 - k_1), \\ X_1 &= \alpha_1 + R'_1x_1 = \alpha_2 + R_2x_2, & Y_1 &= \beta_1 + R'_1y_1 = \beta_2 + R_2y_2, & R'_1 &= \frac{1}{2}(X_1^2 + Y_1^2 - k_1), & R_2 &= \frac{1}{2}(X_1^2 + Y_1^2 - k_2), \\ X_2 &= \alpha_2 + R'_2x_2 = \alpha_3 + R_3x_3, & Y_2 &= \beta_2 + R'_2y_2 = \beta_3 + R_3y_3, & R'_2 &= \frac{1}{2}(X_2^2 + Y_2^2 - k_2), & R_3 &= \frac{1}{2}(X_2^2 + Y_2^2 - k_3), \\ X_3 &= \alpha_3 + R'_3x_3 = \alpha + Rx, & Y_3 &= \beta_3 + R'_3y_3 = \beta + Ry, & R'_3 &= \frac{1}{2}(X_3^2 + Y_3^2 - k_3), & R &= \frac{1}{2}(X_3^2 + Y_3^2 - k). \end{aligned}$$

and then from the values of X, Y, R', R , we have

$$\begin{aligned} \alpha - \alpha_1 + R'x - R_1x_1 &= 0, \\ \beta - \beta_1 + R'y - R_1y_1 &= 0, \end{aligned}$$

giving

$$(\beta - \beta_1)(x - x_1) - (\alpha - \alpha_1)(y - y_1) + (\theta - \theta_1)(xy_1 - x_1y) = 0;$$

and similarly

$$\begin{aligned} (\beta_1 - \beta_2)(x_1 - x_2) - (\alpha_1 - \alpha_2)(y_1 - y_2) + (\theta_1 - \theta_2)(x_1y_2 - x_2y_1) &= 0, \\ (\beta_2 - \beta_3)(x_2 - x_3) - (\alpha_2 - \alpha_3)(y_2 - y_3) + (\theta_2 - \theta_3)(x_2y_3 - x_3y_2) &= 0, \\ (\beta_3 - \beta)(x_3 - x) - (\alpha_3 - \alpha)(y_3 - y) + (\theta_3 - \theta)(x_3y - xy_3) &= 0, \end{aligned}$$

which are the relations connecting the parameters $(x, y), (x_1, y_1), (x_2, y_2), (x_3, y_3)$ of the quadrilateral.

19. We have thus apparently four equations for the determination of four quantities, or the number of quadrilaterals would be finite; but if from the first and second equations we eliminate (x_1, y_1) , and if from the third and fourth equations we eliminate (x_3, y_3) , we find in each case the same relation between $(x, y), (x_2, y_2)$, viz. this is found to be

$$\Omega\Omega_2 = (1 - \alpha x_2 - \beta y_2)^2 (1 - \alpha_2 x - \beta_2 y)^2;$$

and we have thus the singly infinite series of quadrilaterals. We have, of course, between $(x_1, y_1), (x_3, y_3)$ the like relation,

$$\Omega_1\Omega_3 = (1 - \alpha_1 x_3 - \beta_1 y_3)^2 (1 - \alpha_3 x_1 - \beta_3 y_1)^2.$$

20. The relation between $(x, y), (x_1, y_1)$ may be expressed also in the two forms

$$1 - \alpha(x + x_1) - \beta(y + y_1) + (f + \theta_1)xx_1 + (g + \theta_1)yy_1 + \frac{x^2 + y^2}{xy_1 - x_1y} \{(\alpha - \alpha_1)y_1 - (\beta - \beta_1)x_1\} = 0,$$

$$1 - \alpha_1(x + x_1) - \beta_1(y + y_1) + (f + \theta)xx_1 + (g + \theta)yy_1 + \frac{x_1^2 + y_1^2}{xy_1 - x_1y} \{(\alpha_1 - \alpha)y - (\beta_1 - \beta)x\} = 0.$$

In fact, the first of these equations is

$$\begin{aligned} &\{1 + (f + \theta_1)xx_1 + (g + \theta_1)yy_1\}(xy_1 - x_1y) - \{\alpha(x + x_1) + \beta(y + y_1)\}(xy_1 - x_1y) \\ &\quad + \{(\alpha - \alpha_1)y_1 - (\beta - \beta_1)x_1\}(x^2 + y^2) = 0, \end{aligned}$$

which, by virtue of the original form of relation, is

$$\begin{aligned} &\{1 + (f + \theta_1)xx_1 + (g + \theta_1)yy_1\}(xy_1 - x_1y) \\ &\quad - \{\alpha(x + x_1) + \beta(y + y_1)\}(xy_1 - x_1y) + \{(\alpha - \alpha_1)y_1 - (\beta - \beta_1)x_1\}(x^2 + y^2) = 0; \end{aligned}$$

or, in the first term, writing

$$f + \theta_1 = \frac{f - \theta}{f - \theta_1}(f + \theta'_1), \quad g + \theta_1 = \frac{g - \theta}{g - \theta_1}(g + \theta'_1),$$

and in the third term using the like expressions for $\alpha - \alpha_1, \beta - \beta_1$, this is verified.

In this equation the coefficients of α and of β are separately $= 0$: in fact, the coefficient of β is

$$-y(xy_1 - x_1y) - x_1(x^2 + y^2) = -\{x(x_1^2 + y_1^2)\} + x_1(x^2 + y^2) = 0,$$

and similarly the coefficient of α is $= 0$.

And in like manner the second equation may be verified.

21. The two equations are:

$$1 - \alpha x - \beta y - (x^2 + y^2)R' = \alpha x_1 + \beta y_1 - (f + \theta_1)xx_1 - (g + \theta_1)yy_1,$$

$$1 - \alpha_1 x_1 - \beta_1 y_1 - (x_1^2 + y_1^2)R_1 = \alpha_1 x + \beta_1 y - (f + \theta)xx_1 - (g + \theta)yy_1;$$

or, substituting for R' and R_1 their values, these are

$$-\sqrt{\Omega} = -\alpha x_1 - \beta y_1 + (f + \theta_1)xx_1 + (g + \theta_1)yy_1, \quad \sqrt{\Omega_1} = -\alpha_1 x - \beta_1 y + (f + \theta)xx_1 + (g + \theta)yy_1;$$

and similarly

$$\begin{aligned} -\sqrt{\Omega_1} &= -\alpha_1 x_2 - \beta_1 y_2 + (f + \theta_2) x_1 x_2 + (g + \theta_2) y_1 y_2, & \sqrt{\Omega_2} &= -\alpha_2 x_1 - \beta_2 y_1 + (f + \theta_1) x_1 x_2 + (g + \theta_1) y_1 y_2, \\ -\sqrt{\Omega_2} &= -\alpha_2 x_3 - \beta_2 y_3 + (f + \theta_3) x_2 x_3 + (g + \theta_3) y_2 y_3, & \sqrt{\Omega_3} &= -\alpha_3 x_2 - \beta_3 y_2 + (f + \theta_2) x_2 x_3 + (g + \theta_2) y_2 y_3, \\ -\sqrt{\Omega_3} &= -\alpha_3 x_1 - \beta_3 y_1 + (f + \theta_1) x_3 x_1 + (g + \theta_1) y_3 y_1, & -\sqrt{\Omega} &= -\alpha x_3 - \beta y_3 + (f + \theta_3) x_3 x + (g + \theta_3) y_3 y. \end{aligned}$$

Differentiating the equation

$$(\beta - \beta_1)(x - x_1) - (\alpha - \alpha_1)(y - y_1) + (\theta - \theta_1)(xy_1 - x_1y) = 0,$$

we have

$$\begin{aligned} &[(\beta - \beta_1) + (\theta - \theta_1) y_1] dx - [(\alpha - \alpha_1) + (\theta - \theta_1) x_1] dy \\ &- [(\beta - \beta_1) + (\theta - \theta_1) y] dx_1 + [(\alpha - \alpha_1) + (\theta - \theta_1) x] dy_1 = 0; \end{aligned}$$

and writing herein

$$\begin{aligned} dx &= -\frac{(\beta + \theta y)}{\sqrt{\theta}} d\omega, & dy &= \frac{(\alpha + \theta x)}{\sqrt{\theta}} d\omega, \\ dx_1 &= -\frac{(\beta_1 + \theta_1 y_1)}{\sqrt{\theta_1}} d\omega_1, & dy_1 &= \frac{(\alpha_1 + \theta_1 x_1)}{\sqrt{\theta_1}} d\omega_1, \end{aligned}$$

we find

$$\begin{aligned} &-\frac{d\omega}{\sqrt{\theta}} \{(g + \theta)(\beta - \beta_1) y + (f + \theta)(\alpha - \alpha_1) x + (\theta - \theta_1)[(f + \theta) x x_1 + (g + \theta) y y_1]\} \\ &+ \frac{d\omega_1}{\sqrt{\theta_1}} \{(g + \theta_1)(\beta - \beta_1) y_1 + (f + \theta_1)(\alpha - \alpha_1) x_1 + (\theta - \theta_1)[(f + \theta_1) x x_1 + (g + \theta_1) y y_1]\} = 0; \end{aligned}$$

viz., dividing by $\theta - \theta_1$, this becomes

$$-\sqrt{\Omega_1} \frac{d\omega}{\sqrt{\theta}} - \sqrt{\Omega} \frac{d\omega_1}{\sqrt{\theta_1}} = 0, \quad \text{that is,} \quad \frac{d\omega}{\sqrt{\theta}\sqrt{\Omega}} + \frac{d\omega_1}{\sqrt{\theta_1}\sqrt{\Omega_1}} = 0;$$

or, completing the system, we have

$$\frac{d\omega}{\sqrt{\theta}\sqrt{\Omega}} = -\frac{d\omega_1}{\sqrt{\theta_1}\sqrt{\Omega_1}} = \frac{d\omega_2}{\sqrt{\theta_2}\sqrt{\Omega_2}} = -\frac{d\omega_3}{\sqrt{\theta_3}\sqrt{\Omega_3}},$$

which are the differential relations between the parameters $\omega, \omega_1, \omega_2, \omega_3$, or $(x, y), (x_1, y_1), (x_2, y_2), (x_3, y_3)$.

22. From the equations $X = \alpha + R'x, Y = \beta + R'y$, we found

$$\begin{aligned} dX &= \frac{R'd\omega}{(x^2 + y^2)\sqrt{\Omega}} \{Y - (g + \theta) y\}, \\ dY &= \frac{R'd\omega}{(x^2 + y^2)\sqrt{\Omega}} \{X - (f + \theta) x\}; \end{aligned}$$

the new values, $X = \alpha_1 + R_1 x_1$ and $Y = \beta_1 + R_1 y_1$, give in like manner

$$\begin{aligned} dX &= \frac{R_1 d\omega_1}{(x_1^2 + y_1^2)\sqrt{\Omega_1}} \{X - (f + \theta_1) x_1\}, \\ dY &= \frac{R_1 d\omega_1}{(x_1^2 + y_1^2)\sqrt{\Omega_1}} \{Y - (g + \theta_1) y_1\}; \end{aligned}$$

in virtue of the relation just found between $d\omega$ and $d\omega_1$, these two sets of values will agree together if only

$$R'\{Y - (g + \theta)y\} = R_1\{Y - (g + \theta_1)y_1\},$$

$$R'\{X - (f + \theta)x\} = R_1\{X - (f + \theta_1)x_1\}.$$

These are easily verified: the first is

$$R'Y - (g + \theta)(Y - \beta) = (R' - \theta + \theta_1)Y - (g + \theta_1)(Y - \beta_1),$$

viz. this is $(g + \theta)\beta - (g + \theta_1)\beta_1 = 0$, which is right; and similarly the second equation gives $(f + \theta)\alpha - (f + \theta_1)\alpha_1 = 0$, which is right.

From the first values of dX, dY , we have, as above,

$$dS = \frac{eR'S d\omega}{\sqrt{\Omega}\sqrt{\theta}},$$

and the second values give in like manner

$$dS = \frac{e_1R_1S_1 d\omega_1}{\sqrt{\Omega_1}\sqrt{\theta_1}},$$

where e, e_1 is ± 1 . It will be observed that we have in effect, by means of the relation $(\beta - \beta_1)(x - x_1) - (\alpha - \alpha_1)(y - y_1) + (\theta - \theta_1)(xy_1 - x_1y) = 0$, proved the identity of the two values of dS .

Considering the quadrilateral $ABCD$, and giving it an infinitesimal variation, so as to change it into $A'B'C'D'$, then dS is the element of arc AA' ; and writing in like manner dS_1, dS_2, dS_3 for the elements of arc BB', CC', DD' , we have, of course, a like pair of values for each of the elements dS_1, dS_2, dS_3 .

Formulæ for the elements of Arc dS, dS_1, dS_2, dS_3 . Art. Nos. 23 to 27.

23. The formulæ are

$$dS = e'RS \frac{d\omega}{\sqrt{\Omega}\sqrt{\theta}} = e_1R_1S_1 \frac{d\omega_1}{\sqrt{\Omega_1}\sqrt{\theta_1}},$$

$$dS_1 = e'_1R'_1S_1 \frac{d\omega_1}{\sqrt{\Omega_1}\sqrt{\theta_1}} = e_2R_2S_2 \frac{d\omega_2}{\sqrt{\Omega_2}\sqrt{\theta_2}},$$

$$dS_2 = e'_2R'_2S_2 \frac{d\omega_2}{\sqrt{\Omega_2}\sqrt{\theta_2}} = e_3R_3S_3 \frac{d\omega_3}{\sqrt{\Omega_3}\sqrt{\theta_3}},$$

$$dS_3 = e'_3R'_3S_3 \frac{d\omega_3}{\sqrt{\Omega_3}\sqrt{\theta_3}} = eRS \frac{d\omega}{\sqrt{\Omega}\sqrt{\theta}},$$

where the e 's each denote ± 1 . Supposing as above that γ^2 is negative, but that $\gamma'_1, \gamma'_2, \gamma'_3$ are positive, then R, R' have opposite signs; but R'_1, R_1 have the same sign,

as have also R'_2 and R_2 , and R'_3 and R_3 . We may take S, S_1, S_2 , and S_3 as each of them positive: the signs of

$$d\omega, d\omega_1, d\omega_2, d\omega_3 \text{ are } +, -, +, -, \text{ or } -, +, -, +;$$

hence to make dS, dS_1, dS_2, dS_3 all positive,

$$\frac{e}{\sqrt{\theta}\sqrt{\Omega}}, \quad \frac{e'_1}{\sqrt{\Omega_1}\sqrt{\theta_1}}, \quad \frac{e_2}{\sqrt{\Omega_2}\sqrt{\theta_2}}, \quad \frac{e'_3}{\sqrt{\Omega_3}\sqrt{\theta_3}}$$

must have either the signs of

$$R', -R'_1, R'_2, -R'_3; -R, R_1, -R_2, R_3,$$

or else the reverse signs: hence in either case $e' = -e_1 = e'_2 = -e_3$; $e_1 = -e_2 = e'_3 = -e$; or the equations are

$$\begin{aligned} dS &= -eRS \frac{d\omega}{\sqrt{\Omega}\sqrt{\theta}} = e_1R_1S_1 \frac{d\omega_1}{\sqrt{\Omega_1}\sqrt{\theta_1}}, \\ dS_1 &= -e_1R'_1S_1 \frac{d\omega_1}{\sqrt{\Omega_1}\sqrt{\theta_1}} = e_2R_2S_2 \frac{d\omega_2}{\sqrt{\Omega_2}\sqrt{\theta_2}}, \\ dS_2 &= -e_2R'_2S_2 \frac{d\omega_2}{\sqrt{\Omega_2}\sqrt{\theta_2}} = e_3R_3S_3 \frac{d\omega_3}{\sqrt{\Omega_3}\sqrt{\theta_3}}, \\ dS_3 &= -e_3R'_3S_3 \frac{d\omega_3}{\sqrt{\Omega_3}\sqrt{\theta_3}} = eRS \frac{d\omega}{\sqrt{\Omega}\sqrt{\theta}}. \end{aligned}$$

24. But we have $R' - R = \frac{2\sqrt{\Omega}}{x^2+y^2}$, &c.; and hence, putting for shortness

$$\frac{1}{(x^2+y^2)\sqrt{\theta}} = P, \quad \frac{1}{(x_1^2+y_1^2)\sqrt{\theta_1}} = P_1, \quad \frac{1}{(x_2^2+y_2^2)\sqrt{\theta_2}} = P_2, \quad \frac{1}{(x_3^2+y_3^2)\sqrt{\theta_3}} = P_3,$$

and consequently

$$\begin{aligned} dS + dS_3 &= +2ePS d\omega, \\ dS_1 - dS &= -2e_1P_1S_1 d\omega_1, \\ dS_2 - dS_1 &= -2e_2P_2S_2 d\omega_2, \\ dS_3 - dS_2 &= -2e_3P_3S_3 d\omega_3, \end{aligned}$$

$$\begin{aligned} dS &= +ePS d\omega + e_1P_1S_1 d\omega_1 + e_2P_2S_2 d\omega_2 + e_3P_3S_3 d\omega_3, \\ dS_1 &= -ePS d\omega + e_1P_1S_1 d\omega_1 + e_2P_2S_2 d\omega_2 + e_3P_3S_3 d\omega_3, \\ dS_2 &= ePS d\omega - e_1P_1S_1 d\omega_1 - e_2P_2S_2 d\omega_2 + e_3P_3S_3 d\omega_3, \\ dS_3 &= ePS d\omega - e_1P_1S_1 d\omega_1 - e_2P_2S_2 d\omega_2 - e_3P_3S_3 d\omega_3, \end{aligned}$$

which are the required formulæ for the elements of arc.

25. The determination of the signs has been made by means of the particular figure; but it is easy to see that the pairs of terms could not for instance be $dS - dS_3, dS_1 - dS, dS_2 - dS_1, dS_3 - dS$, or any other pairs such that it would be possible to eliminate dS, dS_1, dS_2, dS_3 , and thus obtain an equation such as

$$0 = ePS d\omega + e_1P_1S_1 d\omega_1 + e_2P_2S_2 d\omega_2 + e_3P_3S_3 d\omega_3;$$

this would, by virtue of the relations between $d\omega, d\omega_1, d\omega_2, d\omega_3$, become

$$\frac{eS\sqrt{\Omega}}{x^2 + y^2} - \frac{e_1S_1\sqrt{\Omega_1}}{x_1^2 + y_1^2} + \frac{e_2S_2\sqrt{\Omega_2}}{x_2^2 + y_2^2} - \frac{e_3S_3\sqrt{\Omega_3}}{x_3^2 + y_3^2} = 0,$$

an equation not deducible from the relations which connect $\omega, \omega_1, \omega_2, \omega_3$, and which therefore cannot be satisfied by the variable quadrilateral.

26. The differentials of the formulæ are, it will be observed, of the form $P d\omega$

$$= \frac{S d\omega}{(x^2 + y^2)\sqrt{\theta}},$$

where $\sqrt{\theta} = \sqrt{f + \theta \cdot g + \theta}$ is a mere constant, and

$$x = \frac{\cos \omega}{\sqrt{f + \theta}}, \quad y = \frac{\sin \omega}{\sqrt{g + \theta}},$$

viz. the form is

$$= \frac{\sqrt{(\cos \omega \sqrt{f + \theta} - \alpha)^2 + (\sin \omega \sqrt{g + \theta} - \beta)^2 - \gamma^2}}{\sqrt{\theta} \left(\frac{\cos^2 \omega}{f + \theta} + \frac{\sin^2 \omega}{g + \theta} \right)} d\omega,$$

which is, in fact, the same as Casey's form in ϕ , equation (300), his ϕ being $= 90^\circ - \omega$.

Writing as before v in place of his θ , the differential expression becomes simply $= S dv$; but S^2 expressed as a function of v is an irrational function $M + N\sqrt{V}$, and S would be the root of such a function; so that, if the form originally obtained had been this form $S dv$, it would have been necessary to transform it into the first-mentioned form $\frac{S d\omega}{(x^2 + y^2)\sqrt{\theta}}$, in which S is expressed as a function of (x, y) , that is, of ω .

27. The system of course is

$$\begin{aligned} dS &= eS dv + e_1S_1 dv_1 + e_2S_2 dv_2 + e_3S_3 dv_3, \\ dS_1 &= eS dv - e_1S_1 dv_1 + e_2S_2 dv_2 + e_3S_3 dv_3, \\ dS_2 &= eS dv - e_1S_1 dv_1 - e_2S_2 dv_2 + e_3S_3 dv_3, \\ dS_3 &= eS dv - e_1S_1 dv_1 - e_2S_2 dv_2 - e_3S_3 dv_3, \end{aligned}$$

where $dv = \frac{d\omega}{(x^2 + y^2)\sqrt{\theta}}$, &c.: and this is the most convenient way of writing it.

Reference to Figure. Art. No. 28.

28. I constructed a bicircular quartic consisting of an exterior and interior oval with the following numerical data: $f + \theta_0 = 48$, $f + \theta_1 = 56$, $f + \theta_2 = 60$, $f + \theta_3 = 80$; $g + \theta_0 = -6$, $g + \theta_1 = 2$, $g + \theta_2 = 6$, $g + \theta_3 = 26$,—not very convenient ones, inasmuch as

the exterior oval came out too large. The annexed figure shows 0, 1, 2, 3, the centres of the circles of inversion, the interior oval, and a portion of the exterior oval, also the origin and axes; it will be seen that the centres 0, 2 lie inside the interior oval, the centres 1, 3 outside the exterior oval: I add further the values

$$\begin{array}{llll}
 \sqrt{f + \theta_0} = 6 \cdot 93, & \sqrt{-(g + \theta_0)} = 2 \cdot 44, & \alpha_0 = 10 \cdot 19, & \beta_0 = - \cdot 98, \\
 \sqrt{f + \theta_1} = 7 \cdot 48, & \sqrt{g + \theta_1} = 1 \cdot 41, & \alpha_1 = 8 \cdot 73, & \beta_1 = +2 \cdot 94, \\
 \sqrt{f + \theta_2} = 7 \cdot 75, & \sqrt{g + \theta_2} = 2 \cdot 45, & \alpha_2 = 8 \cdot 15, & \beta_2 = + \cdot 98, \\
 \sqrt{f + \theta_3} = 8 \cdot 94, & \sqrt{g + \theta_3} = 5 \cdot 09, & \alpha_3 = 6 \cdot 10, & \beta_3 = 23.
 \end{array}$$

We thus see how there exists a series of quadrilaterals $ABCD$, where A, B are situate on the interior oval, C, D on the exterior oval. Considering the sides as

drawn in the senses A to B , B to C , C to D , D to A : and representing the inclinations, measured from the positive infinity on the axis of x in the sense x to y , by v_1, v_2, v_3, v respectively: then, in passing to the consecutive quadrilateral $A'B'C'D'$, we have v_1 and v_3 decreasing, v_2 and v increasing, that is, dv_1 and dv_3 negative, dv_2 and dv positive; so that, reckoning the elements AA', BB', CC', DD' , that is, dS, dS_1, dS_2, dS_3 as each of them positive, we have

$$\begin{aligned} dS_2 - dS_1 &= -2S_2 dv_2, \\ dS_3 - dS_2 &= -2S_3 dv_3, \\ dS - dS_3 &= +2S_1 dv_1, \\ dS_1 + dS &= +2S dv, \end{aligned}$$

and thence

$$\begin{aligned} dS &= S dv - S_1 dv_1 - S_2 dv_2 + S_3 dv_3, \\ dS_1 &= S dv + S_1 dv_1 + S_3 dv_3 - S_2 dv_2, \\ dS_2 &= S dv - S_1 dv_1 + S_2 dv_2 - S_3 dv_3, \\ dS_3 &= S dv - S_1 dv_1 - S_2 dv_2 - S_3 dv_3, \end{aligned}$$

which are the correct signs in regard to the particular figure.

$$\text{Reduction of } \int \frac{S d\omega}{(x^2 + y^2)\sqrt{\theta}} \text{ to Elliptic Integrals. Art. No. 29.}$$

29. The expression in question is

$$\int \frac{\sqrt{(\cos \omega \sqrt{f + \theta} - \alpha)^2 + (\sin \omega \sqrt{g + \theta} - \beta)^2 - \gamma^2}}{\sqrt{\theta} \left(\frac{\cos^2 \omega}{f + \theta} + \frac{\sin^2 \omega}{g + \theta} \right)} d\omega,$$

where $\sqrt{\theta}$ is a mere constant; and we may apply it to the Gaussian transformation,

$$\begin{aligned} \cos \omega &= \frac{a + a' \cos T + a'' \sin T}{c + c' \cos T + c'' \sin T}, \\ \sin \omega &= \frac{b + b' \cos T + b'' \sin T}{c + c' \cos T + c'' \sin T}, \end{aligned}$$

where the coefficients $a, b, c, a', b', c', a'', b'', c''$ are such that identically

$$\cos^2 \omega + \sin^2 \omega - 1 = \frac{1}{(c + c' \cos T + c'' \sin T)^2} (\cos^2 T + \sin^2 T - 1);$$

and also

$$\sqrt{(\cos \omega \sqrt{f + \theta} - \alpha)^2 + (\sin \omega \sqrt{g + \theta} - \beta)^2 - \gamma^2} \quad (\text{see formula below}).$$

that is,

$$\begin{aligned} & \cos^2 \omega (f + \theta) + \sin^2 \omega (g + \theta) - 2\alpha\sqrt{f + \theta} \cos \omega - 2\beta\sqrt{g + \theta} \sin \omega + k \\ &= \frac{1}{(c + c' \cos T + c'' \sin T)^2} (\theta_1 - \theta) \{(\theta_1 - \theta) - (\theta_2 - \theta) \cos^2 T - (\theta_3 - \theta) \sin^2 T\}. \end{aligned}$$

30. It is found that G_1, G_2, G_3 are the roots of a cubic equation

$$(G + \theta - \theta_1)(G + \theta - \theta_2)(G + \theta - \theta_3) = 0,$$

which being so, we may assume $G_1 = \theta_1 - \theta$, $G_2 = \theta_2 - \theta$, $G_3 = \theta_3 - \theta$; the second condition, in fact, then is

$$\begin{aligned} & (f + \theta) \cos^2 \omega + (g + \theta) \sin^2 \omega - 2\alpha\sqrt{f + \theta} \cos \omega - 2\beta\sqrt{g + \theta} \sin \omega + k \\ &= \frac{1}{(c + c' \cos T + c'' \sin T)^2} (\theta_1 - \theta) \{(\theta_1 - \theta) - (\theta_2 - \theta) \cos^2 T - (\theta_3 - \theta) \sin^2 T\}; \end{aligned}$$

and this being so, we find without difficulty the values

$$\begin{aligned} a^2 &= \frac{g + \theta_1 \cdot f + \theta_2 \cdot f + \theta_3}{f - g \cdot \theta_1 - \theta_2 \cdot \theta_1 - \theta_3}, & b^2 &= \frac{f + \theta_1 \cdot g + \theta_2 \cdot g + \theta_3}{g - f \cdot \theta_1 - \theta_2 \cdot \theta_1 - \theta_3}, & c^2 &= \frac{f + \theta_1 \cdot g + \theta_1}{\theta_2 - \theta_1 \cdot \theta_3 - \theta_1}, \\ a'^2 &= \frac{g + \theta_2 \cdot f + \theta_1 \cdot f + \theta_3}{f - g \cdot \theta_2 - \theta_1 \cdot \theta_2 - \theta_3}, & b'^2 &= \frac{f + \theta_2 \cdot g + \theta_1 \cdot g + \theta_3}{g - f \cdot \theta_2 - \theta_1 \cdot \theta_2 - \theta_3}, & c'^2 &= \frac{f + \theta_2 \cdot g + \theta_2}{\theta_1 - \theta_2 \cdot \theta_3 - \theta_2}, \\ a''^2 &= \frac{g + \theta_3 \cdot f + \theta_1 \cdot f + \theta_2}{f - g \cdot \theta_3 - \theta_1 \cdot \theta_3 - \theta_2}, & b''^2 &= \frac{f + \theta_3 \cdot g + \theta_1 \cdot g + \theta_2}{g - f \cdot \theta_3 - \theta_1 \cdot \theta_3 - \theta_2}, & c''^2 &= \frac{f + \theta_3 \cdot g + \theta_3}{\theta_1 - \theta_3 \cdot \theta_2 - \theta_3}. \end{aligned}$$

To make these positive, the order of ascending magnitude must, however, be not as heretofore $\theta_0, \theta_1, \theta_2, \theta_3$, but $\theta_0, \theta_2, \theta_1, \theta_3$, viz. we must have $f + \theta_1, f + \theta_2, f + \theta_3, g + \theta_1, g + \theta_2, -(g + \theta_3), \theta_1 - \theta_2, \theta_1 - \theta_3, \theta_2 - \theta_3$ all positive.

31. The above are the values of the squares of the coefficients; we must have definite relations between the signs of the products aa', bb', ab , &c., viz. we may have

$$\begin{aligned} aa'' &= \frac{g + \theta_1}{f - g \cdot \theta_2 - \theta_3} \sqrt{\frac{f + \theta_2 \cdot f + \theta_3}{\theta_3 - \theta_1 \cdot \theta_1 - \theta_2}}, & a'a'' &= \frac{g + \theta_2}{f - g \cdot \theta_3 - \theta_1} \sqrt{\frac{f + \theta_1 \cdot f + \theta_3}{\theta_1 - \theta_2 \cdot \theta_2 - \theta_3}}, \\ b'b'' &= \frac{f + \theta_1}{g - f \cdot \theta_2 - \theta_3} \sqrt{\frac{g + \theta_2 \cdot g + \theta_3}{\theta_3 - \theta_1 \cdot \theta_1 - \theta_2}}, & bb'' &= \frac{f + \theta_2}{g - f \cdot \theta_3 - \theta_1} \sqrt{\frac{g + \theta_1 \cdot g + \theta_3}{\theta_1 - \theta_2 \cdot \theta_2 - \theta_3}}, \\ b'c'' &= \frac{f + \theta_1}{g - f \cdot \theta_1 - \theta_2} \sqrt{\frac{g + \theta_2 \cdot g + \theta_3}{\theta_2 - \theta_3 \cdot \theta_3 - \theta_1}}, \\ b'c &= \frac{g + \theta_1}{g - f \cdot \theta_1 - \theta_2} \sqrt{\frac{f + \theta_2 \cdot f + \theta_3}{\theta_2 - \theta_3 \cdot \theta_3 - \theta_1}}. \end{aligned}$$

and further

$$b'c'' = \frac{f + \theta_2}{f - g \cdot \theta_2 - \theta_3 \cdot \theta_3 - \theta_1} \sqrt{\frac{g + \theta_1 \cdot g + \theta_2 \cdot g + \theta_3}{f - g}}, \quad bc = \frac{f + \theta_1}{\theta_2 - \theta_3 \cdot \theta_1 - \theta_2} \sqrt{\frac{g + \theta_1 \cdot g + \theta_2 \cdot g + \theta_3}{f - g}},$$

$$b''c'' = \frac{f + \theta_3}{f - g \cdot \theta_3 - \theta_1 \cdot \theta_3 - \theta_2} \sqrt{\frac{g + \theta_1 \cdot g + \theta_2 \cdot g + \theta_3}{f - g}}, \quad b''c = \frac{f + \theta_3}{\theta_1 - \theta_2 \cdot \theta_2 - \theta_3} \sqrt{\frac{g + \theta_1 \cdot g + \theta_2 \cdot g + \theta_3}{f - g}},$$

and similarly

$$a'a'' = \frac{g + \theta_1}{\theta_2 - \theta_3 \cdot \theta_3 - \theta_1} \sqrt{\frac{f + \theta_2 \cdot f + \theta_3}{f - g}}, \quad ca' = \frac{g + \theta_1}{\theta_2 - \theta_3 \cdot \theta_3 - \theta_1} \sqrt{\frac{f + \theta_2 \cdot f + \theta_3}{f - g}},$$

$$c''a'' = \frac{g + \theta_2}{\theta_3 - \theta_1 \cdot \theta_1 - \theta_2} \sqrt{\frac{f + \theta_1 \cdot f + \theta_3}{f - g}}, \quad c'a'' = \frac{g + \theta_3}{\theta_1 - \theta_2 \cdot \theta_2 - \theta_3} \sqrt{\frac{f + \theta_1 \cdot f + \theta_2}{f - g}},$$

and also

$$b'c'' + b''c' = \pm \frac{2g + \theta_2 + \theta_3}{\theta_2 - \theta_3} \sqrt{\frac{g + \theta_1 \cdot f + \theta_2 \cdot f + \theta_3}{g - f \cdot \theta_1 - \theta_2 \cdot \theta_1 - \theta_3}},$$

$$= \frac{1}{f + \theta_1} \sqrt{\frac{f + \theta_1 \cdot g + \theta_2 \cdot g + \theta_3}{f - g}} \cdot \frac{2f + \theta_2 + \theta_3}{\theta_2 - \theta_3},$$

$$b''c + bc'' = \pm \frac{2g + \theta_3 + \theta_1}{\theta_3 - \theta_1} \sqrt{\frac{g + \theta_2 \cdot f + \theta_3 \cdot f + \theta_1}{g - f \cdot \theta_2 - \theta_1 \cdot \theta_2 - \theta_3}},$$

$$bc' + b'c = \pm \frac{2g + \theta_1 + \theta_2}{\theta_1 - \theta_2} \sqrt{\frac{g + \theta_3 \cdot f + \theta_1 \cdot f + \theta_2}{g - f \cdot \theta_3 - \theta_1 \cdot \theta_3 - \theta_2}}.$$

32. These values, in fact, satisfy the several relations which exist between the nine coefficients; viz. the original expressions of $\cos \omega, \sin \omega$, in terms of $\cos T, \sin T$ give conversely expressions of $\cos T, \sin T$ in terms of $\cos \omega, \sin \omega$, the two sets being

$$\cos T = \frac{-a + c \cos \omega + c'' a \sin \omega - c''}{a' \cos \omega + b' \sin \omega - c'}, \quad \sin T = \frac{a'' \cos \omega + b'' \sin \omega - c''}{a \cos \omega + b \sin \omega - c};$$

and we have then the relations

$$\cos^2 \omega + \sin^2 \omega - 1 = \frac{1}{(c + c' \cos T + c'' \sin T)^2} (\cos^2 T + \sin^2 T - 1),$$

$$\cos^2 T + \sin^2 T - 1 = \frac{1}{(a \cos \omega + b \sin \omega - c)^2} (\cos^2 \omega + \sin^2 \omega - 1)',$$

$$(\theta + f) \cos^2 \omega + (\theta + g) \sin^2 \omega - 2\alpha \sqrt{\theta + f} \cos \omega - 2\beta \sqrt{\theta + g} \sin \omega + k$$

$$= \frac{1}{(c + c' \cos T + c'' \sin T)^2} (\theta_1 - \theta) \{(\theta_1 - \theta) - (\theta_2 - \theta) \cos^2 T - (\theta_3 - \theta) \sin^2 T\}$$

$$+ \frac{1}{(a \cos \omega + b \sin \omega - c)^2} \{(\theta + f) \cos^2 \omega + (\theta + g) \sin^2 \omega - 2\alpha \sqrt{\theta + f} \cos \omega - 2\beta \sqrt{\theta + g} \sin \omega + k\}.$$

giving the four sets each of six equations

$$\begin{aligned} a^2 + b^2 - c^2 &= -1, & a'a'' + b'b'' - c'c'' &= 0, \\ a'^2 + b'^2 - c'^2 &= +1, & a''a + b''b - c''c &= 0, \\ a''^2 + b''^2 - c''^2 &= +1, & aa' + bb' - cc' &= 0, \\ -a^2 + a'^2 + a''^2 &= +1, & -bc + b'c' + b''c'' &= 0, \\ -b^2 + b'^2 + b''^2 &= +1, & -ca + c'a' + c''a'' &= 0, \\ -c^2 + c'^2 + c''^2 &= -1, & -ab + a'b' + a''b'' &= 0, \end{aligned}$$

$$\begin{aligned} (\theta + f)a^2 + (\theta + g)b^2 - 2\alpha\sqrt{\theta + f}ac - 2\beta\sqrt{\theta + g}bc + kc^2 &= \theta_1 + \theta, \\ (\theta + f)a'^2 + (\theta + g)b'^2 - 2\alpha\sqrt{\theta + f}a'c' - 2\beta\sqrt{\theta + g}b'c' + kc'^2 &= -\theta_2 + \theta, \\ (\theta + f)a''^2 + (\theta + g)b''^2 - 2\alpha\sqrt{\theta + f}a''c'' - 2\beta\sqrt{\theta + g}b''c'' + kc''^2 &= -\theta_3 + \theta, \end{aligned}$$

$$\begin{aligned} (\theta + f)a'a'' + (\theta + g)b'b'' - \alpha\sqrt{\theta + f}(a'c'' + a''c') - \beta\sqrt{\theta + g}(b'c'' + b''c') + kc'c'' &= 0, \\ (\theta + f)a''a + (\theta + g)b''b - \alpha\sqrt{\theta + f}(a''c + ac'') - \beta\sqrt{\theta + g}(b''c + bc'') + kc''c &= 0, \\ (\theta + f)aa' + (\theta + g)bb' - \alpha\sqrt{\theta + f}(ac' + a'c) - \beta\sqrt{\theta + g}(bc' + b'c) + kcc' &= 0, \end{aligned}$$

$$\begin{aligned} (\theta_1 - \theta)a^2 - (\theta_2 - \theta)a'^2 - (\theta_3 - \theta)a''^2 &= \theta + f, \quad \text{or say} \quad (\theta_1 + f)a^2 - (\theta_2 + f)a'^2 - (\theta_3 + f)a''^2 = 0, \\ (\theta_1 - \theta)b^2 - (\theta_2 - \theta)b'^2 - (\theta_3 - \theta)b''^2 &= \theta + g, \quad (\theta_1 + g)b^2 - (\theta_2 + g)b'^2 - (\theta_3 + g)b''^2 = 0, \\ (\theta_1 - \theta)c^2 - (\theta_2 - \theta)c'^2 - (\theta_3 - \theta)c''^2 &= k, \quad \theta_1c^2 - \theta_2c'^2 - \theta_3c''^2 = k + \theta, \\ -(\theta_1 - \theta)bc + (\theta_2 - \theta)b'c' + (\theta_3 - \theta)b''c'' &= -\beta\sqrt{\theta + g}, \\ -(\theta_1 - \theta)ca + (\theta_2 - \theta)c'a' + (\theta_3 - \theta)c''a'' &= -\alpha\sqrt{\theta + f}, \\ -(\theta_1 - \theta)ab + (\theta_2 - \theta)a'b' + (\theta_3 - \theta)a''b'' &= 0; \end{aligned}$$

all which formulæ are in fact satisfied by the foregoing values of the expressions a^2, b^2, a'^2 , &c.

33. We then have

$$d\omega = \frac{dT}{c + c' \cos T + c'' \sin T},$$

the radical which multiplies $d\omega$ being

$$= \frac{1}{c + c' \cos T + c'' \sin T} \sqrt{(\theta_1 - \theta)\{(\theta_1 - \theta) - (\theta_2 - \theta) \cos^2 T - (\theta_3 - \theta) \sin^2 T\}};$$

the differential becomes

$$\frac{dT \sqrt{\theta_1 - \theta_2 \cos^2 T - \theta_3 \sin^2 T}}{\sqrt{\theta} \left(\frac{\cos^2 \omega}{f + \theta} + \frac{\sin^2 \omega}{g + \theta} \right) (c + c' \cos T + c'' \sin T)^3},$$

that is,

$$= \frac{dT \sqrt{\theta_1 - \theta_2 \cos^2 T - \theta_3 \sin^2 T}}{\frac{1}{f + \theta} (a + a' \cos T + a'' \sin T)^2 + \frac{1}{g + \theta} (b + b' \cos T + b'' \sin T)^2}.$$

The denominator could, of course, be reduced to the form $(*)(1, \cos T, \sin T)^2$; but the actual form seems preferable, inasmuch as it puts in evidence the linear factors

$$\frac{1}{\sqrt{f + \theta}} (a + a' \cos T + a'' \sin T) \pm \frac{1}{\sqrt{g + \theta}} (b + b' \cos T + b'' \sin T),$$

and there seems to be no advantage in further reducing the integral.

668.

ON COMPOUND COMBINATIONS.

[From the *Proceedings of the Lit. Phil. Soc. Manchester*, t. XVI. (1877), pp. 113, 114;
Memoirs, ib., Ser. III., t. VI. (1879), pp. 99, 100.]

Prof. Clifford's paper, "On the Types of Compound Statement involving Four Classes," [volume of *Proceedings* quoted, pp. 88–101; *Mathematical Papers*, pp. 1–13], relates mathematically to a question of compound combinations; and it is worth while to consider its connexion with another question of compound combinations, the application of which is a very different one.

Starting with four symbols, A, B, C, D , we have sixteen combinations of the five types 1, $A, AB, ABC, ABCD$, ($1 + 4 + 6 + 4 + 1 = 16$ as before). But in Prof. Clifford's question 1 means $A'B'C'D'$, A means $AB'C'D'$, &c.; viz. each of the symbols means an aggregate of four assertions; and the 16 symbols are thus all of the same type. Considering them in this point of view, the question is as to the number of types of the binary, ternary, &c., combinations of the sixteen combinations; for, according as these are combined,

No. of types of =	1,	2,	3,	4,	5,	6,	7,	8,	9,	10,	11,	12,	13,	14,	15
	1,	4,	6,	19,	27,	47,	55,	78,	55,	47,	27,	19,	6,	4,	1

together.

In the first mentioned point of view the like question arises, in regard to the sets belonging to the five different types separately or in combination with each other; for instance, taking only the six symbols of the type AB , these may be taken 1, 2, 3, 4, or 5 together, and we have in these cases respectively

$$\begin{aligned} \text{No. of types} &= 1, 2, 3, 4, 5, \\ &1, 2, 2, 2, 1. \end{aligned}$$

as is very easily verified; but if the number of letters $A, B, \dots$ be greater (say this = 8), or, instead of letters, writing the numbers 1, 2, 3, 4, 5, 6, 7, 8, then the question is that of the number of types of combination of the 28 duads 12, 13, $\dots$, 78, taken 1, 2, 3, $\dots$, 27 together, a question presenting itself in geometry in regard to the bitangents of a quartic curve (see Salmon's *Higher Plane Curves*, Ed. 2 (1873), pp. 222 *et seq.*): the numbers, so far as they have been obtained, are

No. of types = 1, 2, 3, 4, $\dots$, 24, 25, 26, 27

1, 2, 5, 11, $\dots$, 11, 5, 2, 1.

It might be interesting to complete the series, and, more generally, to determine the number of the types of combination of the $\frac{1}{2}n(n-1)$ duads of n letters.

669.

ON A PROBLEM OF ARRANGEMENTS.

[From the *Proceedings of the Royal Society of Edinburgh*, t. IX. (1878), pp. 338–342.]

It is a well-known problem to find for n letters the number of the arrangements in which no letter occupies its original place; and the solution of it is given by the following general theorem: viz., the number of the arrangements which satisfy any r conditions is

$$\begin{aligned} & (1-1)(1-2)\cdots(1-r), \\ & = 1 - \Sigma(1) + \Sigma(12) - \cdots + (12\dots r), \end{aligned}$$

where 1 denotes the whole number of arrangements; (1) the number of them which fail in regard to the first condition; (2) the number which fail in regard to the second condition; (12) the number which fail in regard to the first condition, and also in regard to the second condition; and so on: $\Sigma(1)$ means $(1) + (2) + \dots + (r)$; $\Sigma(12)$ means $(12) + (13) + \dots + (2r) + \dots + (r-1, r)$; and so on, up to $(12\dots r)$, which denotes the number failing in regard to each of the r conditions.

Thus, in the special problem, the first condition is that the letter in the first place shall not be a ; the second condition is that the letter in the second place shall not be b ; and so on; taking $r = n$, we have the known result,

$$\begin{aligned} \text{No.} &= n^n - \frac{n}{1}(n-1)^n + \frac{n(n-1)}{1\cdot 2}(n-2)^n - \cdots \pm \frac{n!}{n!} \\ &= 1 \cdot 2 \cdot 3 \cdots n \left\{ 1 - 1 + \frac{1}{1\cdot 2} - \frac{1}{1\cdot 2\cdot 3} + \cdots \pm \frac{1}{1\cdot 2 \cdots n} \right\}, \end{aligned}$$

giving for the several cases

$$n = 2, 3, 4, 5, 6, 7, \dots$$

$$\text{No.} = 1, 2, 9, 44, 265, 1854, \dots$$

I proceed to consider the following problem, suggested to me by Professor Tait, in connexion with his theory of knots: to find the number of the arrangements of n letters $abc\dots jk$, when the letter in the first place is not a or b , the letter in the second place not b or c , ..., the letter in the last place not k or a .

Numbering the conditions $1, 2, 3, \dots, n$, according to the places to which they relate, a single condition is called $[1]$; two conditions are called $[2]$ or $[1, 1]$, according as the numbers are consecutive or non-consecutive: three conditions are called $[3]$, $[2, 1]$, or $[1, 1, 1]$, according as the numbers are all three consecutive, two consecutive and one not consecutive, or all non-consecutive; and so on: the numbers which refer to the conditions being always written in their natural order, and it being understood that they follow each other cyclically, so that 1 is consecutive to n . Thus, $n = 6$, the set 125 of conditions is $[3]$, as consisting of 3 consecutive conditions; and similarly 1346 is $[2, 2]$.

Consider a single condition $[1]$, say this is 1 ; the arrangements which fail in regard to this condition are those which contain in the first place a or b ; whichever it be, the other $n - 1$ letters may be arranged in any form whatever; and there are thus $2\Pi(n - 1)$ failing arrangements.

Next for two conditions; these may be $[2]$, say the conditions are 1 and 2 : or else $[1, 1]$, say they are 1 and 3 . In the former case, the arrangements which fail are those which contain in the first and second places ab, ac , or bc : and for each of these, the other $n - 2$ letters may be arranged in any order whatever; there are thus $3\Pi(n - 2)$ failing arrangements. In the latter case, the failing arrangements have in the first place a or b , and in the third place c or d , viz. the letters in these two places are $a.c, a.d, b.c$, or $b.d$, and in each case the other $n - 2$ letters may be arranged in any order whatever: the number of failing arrangements is thus $= 2 \cdot 2 \cdot \Pi(n - 2)$.

And so, in general, when the conditions are $[\alpha, \beta, \gamma, \dots]$, the number of failing arrangements is

$$= (\alpha + 1)(\beta + 1)(\gamma + 1) \cdots \Pi(n - \alpha - \beta - \gamma \cdots).$$

But for $[n]$, that is, for the entire system of the n conditions, the number of failing arrangements is (not as by the rule it should be $= n + 1$, but) $= 2$, viz. the only arrangements which fail in regard to each of the n conditions are (as is at once seen), $abc \dots jk$, and $bc \dots jka$.

Changing now the notation so that $[1], [2], [1, 1]$, &c., shall denote the *number* of the conditions $[1], [2], [1, 1]$, &c., respectively, it is easy to see the form of the general result. If, for greater clearness, we write $n = 6$, we have

$$\begin{aligned} & 1 - \Sigma(1) + \Sigma(12) - \Sigma(123) \\ \text{No.} = & 720 - \{([1] = 6) 2\} 120 + \left\{ \begin{array}{l} ([2] = 6) 3 \\ + ([1, 1] = 9) 2.2 \end{array} \right\} 24 - \left\{ \begin{array}{l} ([3] = 6) 4 \\ + ([2, 1] = 12) 3.2 \\ + ([1, 1, 1] = 2) 2.2.2 \end{array} \right\} 6 \\ & + \Sigma(1234) - \Sigma(12345) + (123456) \\ & + \left\{ \begin{array}{l} ([4] = 6) 5 \\ + ([3, 1] = 6) 4.2 \\ + ([2, 2] = 3) 3.3 \end{array} \right\} 2 - \{([5] = 6) 6\} 1 + \{([6] = 1) 2\}; \end{aligned}$$

or, reducing into numbers, this is

$$\text{No.} = 720 - 1440 + 1296 - 672 + 210 - 36 + 2, = 80.$$

The formula for the next succeeding case, $n = 7$, gives

$$\text{No.} = 5040 - 10080 + 9240 - 5040 + 1764 - 392 + 49 - 2, = 579.$$

Those for the preceding cases, $n = 3, 4, 5$, respectively are

$$\begin{aligned} \text{No.} &= 6 - 12 + 9 - 2 &&= 1, \\ \text{No.} &= 24 - 48 + 40 - 16 + 2 &&= 2, \\ \text{No.} &= 120 - 240 + 210 - 100 + 25 - 2 &&= 13. \end{aligned}$$

We have in general $[1] = n$, $[2] = n$, $[1, 1] = \frac{1}{2}n(n-3)$; and in the several columns of the formulæ the sums of the numbers thus represented are equal to the coefficients of $(1+1)^n$: thus, when $n = 6$ as above, the sums are 6, 15, 20, 15, 6, 1. As regards the calculation of the numbers in question, any symbol $[\alpha, \beta, \gamma]$ is a sum of symbols

$$[\alpha - \alpha' + \beta - \beta' + \gamma - \gamma' + \dots],$$

where $\alpha' + \beta' + \gamma' + \dots$ is any partition of $n - (\alpha + \beta + \gamma + \dots)$; read, of the series of numbers 1, 2, 3, ..., n , taken in cyclical order beginning with any number, retain α , omit α' , retain β , omit β' , retain γ , omit γ' , ... Thus in particular, $n = 6$, $[1, 1]$ is a sum of symbols $[1 - 3 + 1 - 1]$ and $[1 - 2 + 1 - 2]$; it is clear that any such symbol $[\alpha - \alpha' + \beta - \beta' + \dots]$ is $= n$ or a submultiple of n (in particular, if n be prime, the symbol is always $= n$): and we thus in every case obtain the value of $[\alpha, \beta, \gamma, \dots]$ by taking for the negative numbers the several partitions of $n - (\alpha + \beta + \gamma + \dots)$, and for each symbol

$$[\alpha - \alpha' + \beta - \beta' + \gamma - \gamma' + \dots],$$

writing its value, n or a given submultiple of n , as just mentioned. There would, I think, be no use in pursuing the matter further, by seeking to obtain an analytical expression for the symbols $[\alpha, \beta, \gamma, \dots]$.

For the actual formation of the required arrangements, it is of course easy, when all the arrangements are written down, to strike out those which do not satisfy the prescribed conditions, and so obtain the system in question. Or introducing the notion of substitutions¹, and accordingly considering each arrangement as derived by a substitution from the primitive arrangement $abcd \dots jk$, we can write down the substitutions which give the system of arrangements in which no letter occupies its original place: viz. we must for this purpose partition the n letters into parts, no part less than 2, and then in each set taking one letter (say the first in alphabetical order) as fixed, permute in every possible way the other letters of the set; we thus obtain

¹In explanation of the notation of substitutions, observe that $(abcde)$ means that a is to be changed into b , b into c , c into d , d into e , e into a ; and similarly $(ab)(cde)$ means that a is to be changed into b , b into a , c into d , d into e , e into c .

all the substitutions which move every letter. Thus when $n = 5$, we obtain the 44 substitutions for the letters $abcde$, viz. these are

$(abcde)$, &c., 24 symbols obtained by permuting in every way the four letters b, c, d, e ;

$(ab)(cde)$, &c., 20 symbols corresponding to the 10 partitions ab, cde , and for each of them 2 arrangements such as cde, ced .

And then if we reject those symbols which contain in any () two consecutive letters, we have the substitutions which give the arrangements wherein the letter in the first place is not a or b , that in the second place not b or c , and so on. In particular, when $n = 5$, rejecting the substitutions which contain in any (), ab, bc, cd, de , or ea , we have 13 substitutions, which may be thus arranged:

$$\begin{array}{lllll} (acbed), & (ac)(bed), & (acebd), & (adbec), & (aedbc), \\ (aebdc), & (bd)(aec), & & & \\ (acedb), & (ce)(adb), & & & \\ (aecbd), & (ad)(bce), & & & \\ (adceb), & (be)(acd), & & & \end{array}$$

Here in the first column, performing on the symbol $(acbed)$ the substitution $(abcde)$, we obtain $(bdcae)$, $= (aebdc)$, the second symbol; and so again and again operating with $(abcde)$, we obtain the remaining symbols of the column; these are for this reason said to be of the same type. In like manner, symbols of the second column are of the same type; but the symbols in the remaining three columns are each of them a type by itself; viz. operating with $(abcde)$ upon $(acebd)$, we obtain $(bdace)$, $= (acebd)$; and the like as regards $(adbec)$ and $(aedbc)$ respectively. The 13 substitutions are thus of 5 different types, or say the arrangements to which they belong, viz.

$$\begin{array}{lllll} cebad, & ceabd, & cdeab, & deabc, & eabcd, \\ edacb, & edabc, & & & \\ caebd, & daebc, & & & \\ edbac, & debac, & & & \\ daecb, & deacb, & & & \end{array}$$

are of 5 different types. The question to determine for any value of n , the number of the different types, is, it would appear, a difficult one, and I do not at present enter upon it.

670.

[NOTE ON MR MUIR'S SOLUTION OF A "PROBLEM OF
ARRANGEMENT."]

[From the *Proceedings of the Royal Society of Edinburgh*, t. IX. (1878), pp. 388–391.]

The investigation may be carried further: writing for shortness u_3, u_4 , &c., in place of $\Psi(3), \Psi(4)$, &c., the equations are

$$\begin{aligned} v_3 &= 1, \\ v_4 &= 2u_3, \\ v_5 &= 2u_4 + 6u_3 + 1, \\ v_6 &= 4u_5 + 8u_4 + 12u_3, \\ v_7 &= 5u_6 + 10u_5 + 15u_4 + 18u_3 + 1. \end{aligned}$$

Hence assuming

$$u = u_3 + u_4x + u_5x^2 + u_6x^3 + u_7x^4 + \dots,$$

we have

$$\begin{aligned} v &= \frac{1}{1-x^2} + u_3(2x + 12x^2 + 15x^3 + \dots) \\ &\quad + u_4(3x^2 + 8x^3 + 15x^4 + 22x^5 + \dots) \\ &\quad + u_5(4x^3 + 10x^4 + 18x^5 + 26x^6 + \dots) \\ &\quad + u_6(5x^4 + 12x^5 + 21x^6 + 30x^7 + \dots); \end{aligned}$$

so that, forming the equation

$$\begin{aligned} u' - \frac{u}{(1-x)^2} &= u_3(x^2 + 2x^3 + 3x^4 + 4x^5 + \dots) \\ &\quad + u_4(2x^3 + 4x^4 + 6x^5 + 8x^6 + \dots) \\ &\quad + u_5(3x^4 + 6x^5 + 9x^6 + 12x^7 + \dots), \end{aligned}$$

where u' denotes $\frac{du}{dx}$, we have

$$u' - \frac{u}{(1-x)^2} = \frac{1}{1-x}u + \frac{2x}{(1-x)^2(1+x)}u + \frac{2x^3}{(1-x)^2(1+x)}u + \dots$$

or, what is the same thing,

$$u' = \left\{ \frac{1}{(1-x)^2} + \frac{2x}{(1-x)^2(1+x)} \right\} u - \frac{x^2}{(1-x)^2(1+x)} + \frac{1}{1-x^2}.$$

This equation may be simplified: write

$$u = \frac{1}{1-x^2}Q - \left(-\frac{1}{x} + \frac{1}{2} \right) Q,$$

then

$$u' = \left(-\frac{1}{x^2} - \frac{1}{2} \right) Q + \frac{1-x^2}{x^2} Q',$$

and the equation is

$$\left\{ -\frac{1-x^2}{x^2} + \frac{2}{x^2} \frac{1+x}{1+x} - \frac{2x^2}{x^2(1-x)} - \frac{2}{x^2} \frac{1}{x(1+x)} \right\} Q + \frac{1+x}{1-x^2} Q' = \frac{1-x}{1+x},$$

that is,

$$\left\{ -\frac{1}{x^2} + \frac{2}{x^3} \right\} Q + \frac{1+x^2}{x^2} Q' = \frac{1-x}{1+x},$$

or

$$\frac{-(1-x^2)}{x^4} Q + \frac{1-x^2}{x^2} Q' = \frac{1-x}{1+x},$$

or finally,

$$Q' \left(1 - \frac{1}{x^2} \right) - \frac{1}{x^4} Q = \frac{1-x}{1+x},$$

giving

$$Q \left(1 - \frac{1}{x^2} \right) - Q' \cdot \frac{1}{x^2} = \frac{1-x}{1+x},$$

and thence

$$Q = e^{(x-x^{-1})} \int e^{-(x-x^{-1})} \frac{x^2}{(x+1)^2} dx,$$

which is the value of the generating function

$$u = u_3 + u_4x + u_5x^2 + \&c.$$

But for the purpose of calculation it is best to integrate by a series the differential equation for Q : assuming

$$Q = q_3 + q_4x + q_5x^2 + \&c.,$$

we find

$$\begin{aligned} q_4 &= 4q_3 &= 1, \\ q_5 &= 5q_4 + q_3 &= -2, \\ q_6 &= 6q_5 + q_4 &= +3, \\ q_7 &= 7q_6 + q_5 &= -4, \\ q_8 &= 8q_7 + q_6 &= +5, \\ &\vdots \\ q_n &= nq_{n-1} + q_{n-2} + (-)^{n-1}(n-2). \end{aligned}$$

We have thus for $q_3, q_4, q_5, \dots$ the values 1, 2, 14, 82, 593, 4820, $\dots$, and thence viz. writing

$$u = (1 - x^2)Q = (1 + x)(1 + 2x + 14x^2 + 82x^3 + 593x^4 + 4820x^5 + \dots),$$

1	2	14	82	593	4820...
-1	-2	-14	-82		

the values of $u_3, u_4, \dots$ are 1, 2, 13, 80, 579, 4738, $\dots$, agreeing with the results found above.

In the more simple problem, where the arrangements of the n things are such that no one of them occupies its original place, if u_n be the number of arrangements, we have

$$\begin{aligned} u_2 &= 1 &= 1, \\ u_3 &= 2u_2 &= 2, \\ u_4 &= 3u_3 + (u_3 - u_2) &= 9, \\ u_5 &= 4u_4 + (u_4 - u_3) &= 44, \\ u_{n+1} &= nu_n + (u_n - u_{n-1}), \end{aligned}$$

and writing

$$u = u_2 + u_3x + u_4x^2 + \dots,$$

we find

$$1 + (2x + 3x^2 + \dots)u + (x^2 + x^3)u' = u(-1 + 2x + 3x^2 + \dots)u + (x^2 + x^3)u' = 1,$$

that is,

$$u(-1 + 2x + 3x^2) + (x^2 + x^3)u' = 1,$$

or, what is the same thing,

$$\frac{u}{x^2} + \left(\frac{3}{x} - 1\right)u' = \frac{1}{x^2},$$

whence

$$u = \frac{1}{x^2} e^{-\frac{1}{x}} \int e^{\frac{1}{x}} \frac{1}{1+x} dx.$$

But the calculation is most easily performed by means of the foregoing equation of differences, itself obtained from the differential equation written in the foregoing form,

$$u(-1 + 2x + 3x^2) + (x^2 + x^3)u' = 1.$$

671.

ON A SIBI-RECIPROCAL SURFACE.

[From the *Berlin. Akad. Monatsber.*, (1878), pp. 309–313.]

The question of the generation of a sibi-reciprocal surface—that is, a surface the reciprocal of which is of the same order and has the same singularities as the original surface—was considered by me in the year 1868, see *Proc. London Math. Soc.* t. II. pp. 61–63, [part of 387], where it is remarked that if a surface be considered as the envelope of a quadric surface varying according to given conditions, then the reciprocal surface is given as the envelope of a quadric surface varying according to the reciprocal conditions; whence, if the conditions be sibi-reciprocal, it follows that the surface is a sibi-reciprocal surface. And I gave as instances the surface which is the envelope of a quadric surface touching each of 8 given lines; and also the surface called the “tetrahedroid,” which is a homographic transformation of Fresnel’s Wave Surface and a particular case of the quartic surface with 16 nodes.

The interesting surface of the order 8, recently considered by Herr Kummer, *Berl. Monatsber.*, Jan. 1878, pp. 25–36, is included under the theory. In fact, if we consider a line L , whereof the six coordinates

$$a, b, c, f, g, h,$$

satisfy each of the three linear relations

$$\begin{aligned} f_1a + g_1b + h_1c + a_1f + b_1g + c_1h &= 0, \\ f_2a + g_2b + h_2c + a_2f + b_2g + c_2h &= 0, \\ f_3a + g_3b + h_3c + a_3f + b_3g + c_3h &= 0, \end{aligned}$$

the locus of this line is a quadric surface the equation of which is

$$\begin{aligned} T &= (agh)x^2 + (bhf)y^2 + (cfg)z^2 + (abc)w^2 \\ &+ [(abg) - (cah)]xw + [(bfg) + (chf)]yz \\ &+ [(bch) - (abf)]yw + [(cgh) - (afg)]zx \\ &+ [(caf) - (bcg)]zw + [(ahf) + (bgh)]xy = 0. \end{aligned}$$

where (agh) is used to denote the determinant $\begin{vmatrix} a_1 & g_1 & h_1 \\ a_2 & g_2 & h_2 \\ a_3 & g_3 & h_3 \end{vmatrix}$, and so for the other symbols.

Considering the reciprocal of the line L in regard to the quadric surface $X^2 + Y^2 + Z^2 + W^2 = 0$, the six coordinates of the reciprocal line are

$$f, g, h, a, b, c,$$

and it is hence at once seen that the locus of the reciprocal line is the quadric surface $T' = 0$ obtained from the equation by interchanging therein the symbolical quantities a, b, c and f, g, h : viz. writing also $(\xi, \eta, \zeta, \omega)$ in place of (x, y, z, w) , the new equation is

$$\begin{aligned} T' = & (fbc)\xi^2 + (gca)\eta^2 + (hab)\zeta^2 + (fgh)\omega^2 \\ & + [(fgh) - (hfc)]\zeta\omega + [(fab) + (hca)]\eta\zeta \\ & + [(ghc) - (fga)]\eta\omega + [(gbc) + (fab)]\xi\omega \\ & + [(hbc) - (ghb)]\xi\zeta + [(hca) + (gbc)]\xi\eta = 0; \end{aligned}$$

or, what is the same thing, this equation $T' = 0$ is the equation of the original quadric surface (the locus of L) expressed in terms of the plane-coordinates ξ, η, ζ, ω .

Now considering each of the quantities $a_1, b_1, c_1, f_1, g_1, h_1, a_2, b_2$, etc., a_3, b_3 , etc., as a given linear function of a variable parameter λ , say $a_1 = a'_1 + a''_1\lambda$, $b_1 = b'_1 + b''_1\lambda$, etc., the equation $T = 0$ takes the form

$$A\lambda^3 + 3B\lambda^2 + 3C\lambda + D = 0,$$

where A, B, C, D are given quadric functions of the coordinates x, y, z, w ; and the envelope of the quadric surface $T = 0$ is Herr Kummer's surface of the eighth order

$$(AD - BC)^2 - 4(AC - B^2)(BD - C^2) = 0.$$

In like manner the equation $T' = 0$ takes the form

$$A'\lambda^3 + 3B'\lambda^2 + 3C'\lambda + D' = 0,$$

where A', B', C', D' are given functions of the coordinates ξ, η, ζ, ω ; and we have

$$(A'D' - B'C')^2 - 4(A'C' - B'^2)(B'D' - C'^2) = 0,$$

as the equation of the reciprocal surface; or (what is the same thing) as that of the original surface, regarding ξ, η, ζ, ω as plane-coordinates.

In regard to the foregoing equation $T = 0$, it is to be noticed that, if $a_1, b_1, c_1, f_1, g_1, h_1, a_2, b_2$, etc., a_3, b_3 , etc., instead of being arbitrary coefficients, were the coordinates of three given lines L_1, L_2, L_3 respectively; that is, if we had

$$\begin{aligned} a_1f_1 + b_1g_1 + c_1h_1 &= 0, \\ a_2f_2 + b_2g_2 + c_2h_2 &= 0, \\ a_3f_3 + b_3g_3 + c_3h_3 &= 0, \end{aligned}$$

then the three linear relations satisfied by (a, b, c, f, g, h) would express that the line L was a line meeting each of the three given lines L_1, L_2, L_3 : the locus is therefore the quadric surface which passes through these three lines; and I have in my paper “On the six coordinates of a Line,” *Camb. Phil. Trans.*, t. XI. (1869), pp. 290–323, [435], found the equation to be the foregoing equation $T = 0$. But it is easy to see that the same equation subsists in the case where the three equations $a_1f_1 + b_1g_1 + c_1h_1 = 0$, etc., are not satisfied. For the several coefficients being perfectly general, any one of the three linear relations may be replaced by a linear combination of these equations; that is, in place of $a_1, b_1, c_1, f_1, g_1, h_1$, we may write $a'_1, b'_1, c'_1, f'_1, g'_1, h'_1$, where $a'_1 = \theta_1a_1 + \theta_2a_2 + \theta_3a_3$, $b'_1 = \theta_1b_1 + \theta_2b_2 + \theta_3b_3$, etc.; and these factors $\theta_1, \theta_2, \theta_3$ may be conceived to be such that the condition in question $a'_1f'_1 + b'_1g'_1 + c'_1h'_1 = 0$ is satisfied. Similarly the second set of coefficients may be replaced by $a'_2, b'_2, c'_2, f'_2, g'_2, h'_2$, where $a'_2 = \phi_1a_1 + \phi_2a_2 + \phi_3a_3$, etc., and the condition $a'_2f'_2 + b'_2g'_2 + c'_2h'_2 = 0$ is satisfied: and the third set by $a'_3, b'_3, c'_3, f'_3, g'_3, h'_3$, where $a'_3 = \psi_1a_1 + \psi_2a_2 + \psi_3a_3$, etc., and the condition $a'_3f'_3 + b'_3g'_3 + c'_3h'_3 = 0$ is satisfied. We have therefore an equation $T' = (a'g'h')x^2 + \text{etc.}$, which only differs from the equation $T = 0$ by having therein the accented letters in place of the unaccented ones: and, substituting for the accented letters their values, the whole divides by the determinant $(\theta\phi\psi)$, and throwing this out we obtain the required equation $T = 0$.

But it is easier to obtain the equation $T = 0$ directly. We have

$$\begin{aligned} hy - gz &+ aw = 0, \\ -hx &+ fz + bw = 0, \\ gx - fy &+ cw = 0, \\ -ax - by - cz &= 0; \end{aligned}$$

viz. in virtue of the equation $af + bg + ch = 0$ which connects the six coordinates, these four equations are equivalent to two independent equations which are the equations of the line (a, b, c, f, g, h) : or, what is the same thing, any three of these equations imply the fourth equation and also the relation $af + bg + ch = 0$.

We might, from the three linear relations and any three of the last-mentioned four equations, eliminate a, b, c, f, g, h and so obtain the required equation $T = 0$; but it is better, introducing the arbitrary coefficients $\alpha, \beta, \gamma, \delta$, to employ all the four equations. The result of the elimination is thus given in the form

$$\begin{vmatrix} \alpha, & w, & -z, & y & & & \\ \beta, & & w, & z, & -x & & \\ \gamma, & & & w, & -y, & x & \\ \delta, & x, & y, & z, & & & \\ & f_1, & g_1, & h_1, & a_1, & b_1, & c_1 \\ & f_2, & g_2, & h_2, & a_2, & b_2, & c_2 \\ & f_3, & g_3, & h_3, & a_3, & b_3, & c_3 \end{vmatrix} = 0,$$

viz. the left-hand side here contains the factor $-(\alpha x + \beta y + \gamma z + \delta w)$; throwing this out, we obtain the required quadric equation $T = 0$. If for the calculation of T we compare the terms containing δ , we have

$$Tw = \begin{vmatrix} w, & -z, & y & & & \\ & w, & z, & -x & & \\ & & w, & -y, & x & \\ f_1, & g_1, & h_1, & a_1, & b_1, & c_1 \\ f_2, & g_2, & h_2, & a_2, & b_2, & c_2 \\ f_3, & g_3, & h_3, & a_3, & b_3, & c_3 \end{vmatrix},$$

where observe that, writing $w = 0$, the right-hand side vanishes as containing the factor

$$\begin{vmatrix} & -z, & y \\ z, & & -x \\ -y, & x, & \end{vmatrix}.$$

Hence the right-hand side divides by w ; and one of its terms being evidently $w^2(abc)$, T contains as it should do the term $(abc)w^2$: the remaining terms can be found without any difficulty, and the foregoing expression for T is thus verified.

672.

ON THE GAME OF MOUSETRAP.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 8–10.]

In the note “A Problem in Permutations,” *Quarterly Mathematical Journal*, t. I. (1857), p. 79, [161], I have spoken of the problem of permutations presented by this game.

A set of cards—ace, two, three, &c., say up to thirteen—are arranged (in any order) in a circle with their faces upwards; you begin at any card, and count one, two, three, &c., and if upon counting, suppose the number five, you arrive at the card five, the card is thrown out; and beginning again with the next card, you count one, two, three, &c., throwing out (if the case happen) a new card as before, and so on until you have counted up to thirteen, without coming to a card which has to be thrown out. The original question proposed was: for any given number of cards to find the arrangement (if any) which would throw out all the cards in a given order; but (instead of this) we may consider all the different arrangements of the cards, and inquire how many of these there are in which all or any given smaller number of the cards will be thrown out; and (in the several cases) in what orders the cards are thrown out. Thus to take the simple case of four cards, the different arrangements, with the cards thrown out in each, are

1, 2, 3, 4	1,
1, 2, 4, 3	1,
1, 3, 2, 4	1, 2, 3, 4,
1, 3, 4, 2	1,
1, 4, 2, 3	1,
1, 4, 3, 2	1,
2, 1, 3, 4	3, 4,
2, 1, 4, 3	— — —
2, 3, 4, 1	— — —
2, 3, 1, 4	4,
2, 4, 1, 3	— — —
2, 4, 3, 1	— — —
3, 1, 2, 4	4,
3, 1, 4, 2	3, 2,
3, 2, 1, 4	— — —
3, 2, 4, 1	4, 2, 1, 3,
3, 4, 1, 2	— — —
3, 4, 2, 1	— — —
4, 1, 2, 3	— — —
4, 1, 3, 2	3,
4, 2, 1, 3	2, 1, 3, 4,
4, 2, 3, 1	3, 1, 2, 4,
4, 3, 1, 2	— — —
4, 3, 2, 1	— — —

Classifying these so as to show in how many arrangements a given card or permutation of cards is thrown out, we have the table

No.	Thrown out
9	none
4	1
1	3
2	4
1	3, 2
1	2, 3
1	3, 4
1	1, 3, 4, 2
1	1, 2, 3, 4
1	4, 2, 1, 3
1	2, 1, 3, 4
1	3, 1, 2, 4,
24	

viz. there are nine arrangements in which no card is thrown out, four arrangements in which only the card 1 is thrown out, one arrangement in which only the card 3 is thrown out, and so on.

It will be observed that there are five arrangements in which all the cards are thrown out, each throwing them out in a different order; there are thus only five orders in which all the cards are thrown out.

The general question is of course to form a like table for the numbers 5, 6, ..., or any greater number of cards.

673.

NOTE ON THE THEORY OF CORRESPONDENCE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 32, 33.]

If the point P on a given curve U of the order m , and the point Q on a given curve V of the order m' , have a $(1, 1)$ correspondence, this implying that the two curves have the same deficiency; then if PQ intersects the consecutive line $P'Q'$ in a point R , the locus of R is a curve W of the class $m' + m$, and the point R on this curve has, in general (but not universally), a $(1, 1)$ correspondence with the point P on U , or with the point Q on V . For, considering the correspondence of the points P and R , to a given position of P there corresponds, it is clear, a single position of R ; on the other hand, starting from R , the tangent at this point to the curve W meets the curve U in m points and the curve V in m' points, but it is in general only one of the m points and only one of the m' points which are corresponding points on the curves U and V ; that is, it is only one of the m points which is a point P ; and the correspondence of (P, R) is thus a $(1, 1)$ correspondence.

But the curves U, V may be such that the correspondence of (P, R) is not a $(1, 1)$ but a $(k, 1)$ correspondence; viz., that to a given position of P there corresponds a single position of R , but to a given position of R , k positions of P . To show that this is so, imagine through P a line Π having therewith a $(k, 1)$ correspondence; P being, as above, a point on the curve U , the line in question envelopes a curve W ; and the correspondence is such that, for any given position of P on the curve U , we have through it a single position of the line: but, for a given tangent of the curve W , we have upon it k positions of the point P , viz. k of the intersections of the line with the curve U are points corresponding to the line; this, of course, implies that the curve U is not any curve whatever of the order m , but a curve of a peculiar nature.

Imagine now that we have on the line Π a point Q , having with P a $(1, 1)$ correspondence of a given nature: to fix the ideas, suppose P, Q are harmonics in regard to a given conic: since on each of the lines Π there are k positions of P , there are also on the line k positions of Q , and the locus of these k points Q is a curve V , say of the order m' .

The point P on the curve U and the point Q on the curve V have a $(1, 1)$ correspondence. For, consider P as given: there is a single position of the line Π intersecting V in m' points, but obviously only one of these is the point Q . And consider Q as given: then through Q we have say μ tangents of the curve W ; each of these tangents intersects the curve U in m points, k of which are points P , but for a tangent taken at random no one of these is the correspondent of Q ; it is, in general, only one of the μ tangents which has upon it k points P , one of them being the point corresponding to Q ; that is, to a given position of Q there corresponds a single position of P ; and the correspondence of the points (P, Q) is thus a $(1, 1)$ correspondence.

We have thus the point P on the curve U and the point Q on the curve V , which points have with each other a $(1, 1)$ correspondence; and the line Π is the line joining these points; this intersects the consecutive line PQ in a point R ; and the locus of R is the curve W . To a given position of P there corresponds a single line Π , and therefore a single position of R ; but to a given position of R there correspond k positions of P , viz. drawing at R the tangent to the curve W , this is a line Π having upon it k points P , or the correspondence of (P, Q) is, as stated, a $(k, 1)$ correspondence.

The foregoing considerations were suggested to me by the theory of parallel curves. Take a curve parallel to a given curve, for example, the ellipse; this is a curve of the order 8, such that every normal thereto is a normal at two distinct points; and the curve has as its evolute the evolute of the ellipse, or, more accurately, the evolute of the ellipse taken *twice*; but, attending only to the evolute taken once, each tangent of the evolute is a normal of the parallel curve at two distinct points thereof, and the points of the parallel curve have with those of the evolute not a $(1, 1)$ but a $(2, 1)$ correspondence.

674.

NOTE ON THE CONSTRUCTION OF CARTESIANS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), p. 34.]

If $\rho = a + b \cos \theta$, and $r = \frac{1}{2}\{\rho \pm \sqrt{(\rho^2 - c^2)}\}$, then obviously $r^2 - r\rho + \frac{1}{4}c^2 = 0$, that is,

$$r^2 - r(a + b \cos \theta) + \frac{1}{4}c^2 = 0,$$

which is the equation of a Cartesian. Here $\rho = a + b \cos \theta$ is the equation of a limaçon or nodal Cartesian, having the origin for the node; and for any given value of θ , deducing from the radius vector of the limaçon the new radius vector r by the above formula $r = \frac{1}{2}\{\rho \pm \sqrt{(\rho^2 - c^2)}\}$, we obtain a Cartesian, or by giving different values to c , a series of Cartesians having the origin for a common focus. The construction is a very convenient one.

675.

ON THE FLEFLECNODAL PLANES OF A SURFACE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 49–51.]

If at a node (or double point) of a plane curve there is on one of the branches an inflexion, (that is, if the tangent has a 3-pointic intersection with the branch), the node is said to be a flecnod; and if there is on each of the branches an inflexion, then the node is said to be a fleflecnode. The tangent plane of a surface intersects the surface in a plane curve having at the point of contact a node; if this is a flecnod or a fleflecnode, the tangent plane is said to be a flecnodal or a fleflecnodal plane accordingly. For a quadric surface each tangent plane is fleflecnodal: this is obvious geometrically (since the section is a pair of lines), and it will presently appear that the analytical condition for such a plane is satisfied. In fact, if the origin be taken at a point of a surface, so that $z = 0$ shall be the equation of the tangent plane, then in the neighbourhood of the point we have

$$z = (x, y)^2 + (x, y)^3 + \&c.;$$

and the condition for a fleflecnodal plane is that the term $(x, y)^2$ shall be a factor of the succeeding term $(x, y)^3$. Now for a quadric surface the equation is

$$z = \frac{1}{2}\{ax^2 + 2hxy + by^2 + 2(fy + gx)z + cz^2\}$$

that is,

$$z(1 - fy - gx - \frac{1}{2}cz) = \frac{1}{2}(ax^2 + 2hxy + by^2),$$

or developing as far as the third order in (x, y) , we have

$$z = \frac{1}{2}(ax^2 + 2hxy + by^2)(1 + fy + gx),$$

so that the condition in question is satisfied.

In what follows, I take for greater simplicity $h = 0$, (viz. $x = 0, y = 0$ are here the tangents to the two curves of curvature at the point in question), and to avoid fractions write $2f, 2g$ in place of f, g respectively; the developed equation of the quadric surface is thus

$$z = \frac{1}{2}(ax^2 + by^2) + (ax^2 + by^2)(gx + fy).$$

I consider the parallel surface, obtained by measuring off on the normal a constant length k . If, as usual, p, q denote $\frac{dz}{dx}$ and $\frac{dz}{dy}$ respectively, then, in general, (X, Y, Z) being the coordinates of the point on the parallel surface,

$$Z = z + \frac{k}{\sqrt{1 + p^2 + q^2}},$$

$$X = x - \frac{kp}{\sqrt{1 + p^2 + q^2}},$$

$$Y = y - \frac{kq}{\sqrt{1 + p^2 + q^2}}.$$

But in the present case

$$p = ax + 3agx^2 + 2afxy + bgy^2,$$

whence

$$q = by + afx^2 + 2bgxy + 3bfy^2,$$

$$X = x - k\{ax + 3agx^2 + 2afxy + bgy^2\},$$

$$Y = y - k\{by + afx^2 + 2bgxy + 3bfy^2\};$$

or, putting for convenience,

$$X = (1 - ka)\xi, \quad Y = (1 - kb)\eta,$$

then, for a first approximation $x = \xi, y = \eta$; whence, writing

$$P = 3ag\xi^2 + 2af\xi\eta + bg\eta^2, \quad Q = af\xi^2 + 2bg\xi\eta + 3bf\eta^2,$$

we find

$$x = \xi + \frac{kP}{1 - ka}, \quad y = \eta + \frac{kQ}{1 - kb},$$

and thence

$$p = a\left(\xi + \frac{kP}{1 - ka}\right) + P = a\xi + \frac{P}{1 - ka},$$

$$q = b\eta + \frac{Q}{1 - kb}.$$

Hence

$$Z = k + \frac{1}{2}\{a(1 - ka)\xi^2 + b(1 - kb)\eta^2\} + (a\xi^2 + b\eta^2)(g\xi + f\eta).$$

where, changing the origin to the point $x = 0, y = 0, z = k$ on the parallel surface, the coordinates of the consecutive point are $Z - k, X = (1 - ka)\xi$, and $Y = (1 - kb)\eta$.

We cannot, by any determination of the value of k , make the plane $Z - k = 0$ a flefnodal plane of the parallel surface; but if

$$\frac{a\xi^2 + b\eta^2}{a^2\xi^2 + b^2\eta^2} = k,$$

then

$$1 - ka = \frac{b\eta^2(a - b)}{a^2\xi^2 + b^2\eta^2}, \quad 1 - kb = \frac{a\xi^2(b - a)}{a^2\xi^2 + b^2\eta^2},$$

and the equation becomes

$$Z - k = \frac{1}{2} \frac{ab(b - a)}{a^2\xi^2 + b^2\eta^2} (\xi^2 - \eta^2)(a^2\xi^2 + b^2\eta^2) (g\xi + f\eta),$$

viz. the term of the second has here a factor $g\xi + f\eta$ which divides the term of the third order, and the equation $Z - k = 0$ is a flecnodal plane of the parallel surface.

676.

NOTE ON A THEOREM IN DETERMINANTS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 55–57.]

It is well known that if 12, &c., denote the determinants formed with the matrix

$$\begin{vmatrix} \alpha, & \beta, & \gamma, & \delta \\ \alpha', & \beta', & \gamma', & \delta' \end{vmatrix},$$

then, identically,

$$12.34 + 13.42 + 14.23 = 0.$$

The proper proof of the theorem is obtained by remarking that we have

$$0 = \begin{vmatrix} \alpha, & \beta, & \gamma, & \delta \\ \alpha', & \beta', & \gamma', & \delta' \\ \alpha, & \beta, & \gamma, & \delta \\ \alpha', & \beta', & \gamma', & \delta' \end{vmatrix},$$

as at once appears by subtracting the first and second lines from the third and fourth lines respectively; and, this being so, the development of the determinant gives the theorem. The theorem might, it is clear, have been obtained in four different forms according as in the determinant the missing terms were taken to be as above (δ, δ') , or to be (α, α') , (β, β') , or (γ, γ') ; but the four results are equivalent to each other.

There is obviously a like theorem for the sums of products of determinants formed with the matrix

$$\begin{vmatrix} \alpha, & \beta, & \gamma, & \delta, & \epsilon, & \zeta \\ \alpha', & \beta', & \gamma', & \delta', & \epsilon', & \zeta' \\ \alpha'', & \beta'', & \gamma'', & \delta'', & \epsilon'', & \zeta'' \end{vmatrix},$$

viz. the theorem is obtained by development of the determinant in an identical equation, such as

$$0 = \begin{vmatrix} \alpha, & \beta, & \gamma, & \delta, & \epsilon, & \zeta \\ \alpha', & \beta', & \gamma', & \delta', & \epsilon', & \zeta' \\ \alpha'', & \beta'', & \gamma'', & \delta'', & \epsilon'', & \zeta'' \\ \alpha, & \beta, & \gamma, & \delta, & \epsilon, & \zeta \\ \alpha', & \beta', & \gamma', & \delta', & \epsilon', & \zeta' \\ \alpha'', & \beta'', & \gamma'', & \delta'', & \epsilon'', & \zeta'' \end{vmatrix},$$

but we thus obtain 15 results which are not all equivalent.

If, for shortness, we write

$$\begin{aligned} A &= 123 \cdot 456, \\ -B &= 124 \cdot 356, \\ -C &= 125 \cdot 346, \\ D &= 126 \cdot 345, \\ -E &= 134 \cdot 256, \\ -F &= 135 \cdot 246, \\ G &= 136 \cdot 245, \\ -H &= 145 \cdot 236, \\ I &= 146 \cdot 235, \\ J &= 156 \cdot 234, \end{aligned}$$

then the fifteen results are

$$\begin{aligned} A + B - C - D &= 0, \\ A + B - E - J &= 0, \\ A - C + F - I &= 0, \\ A - D + G - H &= 0, \\ A - E + F + G &= 0, \\ A - H - I - J &= 0, \\ B - C - G + H &= 0, \\ B - D - F + I &= 0, \\ B - E + H + I &= 0, \\ B - F - G - J &= 0, \\ C + D - E - J &= 0, \\ C - E + G + I &= 0, \\ C - F - H - J &= 0, \\ D - E + F + H &= 0, \\ D - G - I - J &= 0, \end{aligned}$$

which are all satisfied if only

$$\begin{aligned} A &= \dots + H + I + J, \\ B &= F + G + \dots + J, \\ C &= F + \dots + H + J, \\ D &= \dots + G + \dots + J, \\ E &= F + G + H + I + J; \end{aligned}$$

and we thus have these five relations between the ten products of determinants $A, B, C, D, E, F, G, H, I, J$.

677.

[ADDITION TO MR GLAISHER'S PAPER "PROOF OF STIRLING'S THEOREM."]

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 63, 64.]

It is easy to extend Mr Glaisher's investigation so as to obtain from it the more approximate value

$$\Pi n = \sqrt{(2\pi)} n^{n+\frac{1}{2}} e^{-n-\frac{1}{12n}}.$$

We, in fact, have

$$\psi x = e^{ax+bx^2+cx^3+\dots},$$

where $a, b, \dots$ are given functions of n , viz.

$$a = \frac{1}{2} \left\{ \frac{1}{3^2} + \frac{1}{5^2} + \dots + \frac{1}{(2n+1)^2} \right\},$$

$$b = \frac{1}{3} \left\{ \frac{1}{3^3} + \frac{1}{5^3} + \dots + \frac{1}{(2n+1)^3} \right\},$$

&c.

And hence writing $x = 1$, we have

$$\psi(1) = \frac{1}{2} \sum_0^n \frac{1}{(2n+2)^{n+1}} (2n+2)^{n+1} = e^{a+b+c+\dots},$$

that is,

$$\begin{aligned} \Pi n &= \left(\frac{2n+2}{e} \right)^{n+1} e^{-(n+1)+(a+b+\dots)} \\ &= (n+1)^{n+1} e^{-n-1+(a+b+\dots)} \\ &= n^{n+1} \left(1 + \frac{1}{n} \right)^{n+1} e^{-n-1+(a+b+\dots)}. \end{aligned}$$

Hence for $(1 + \frac{1}{n})^{n+1}$ writing $e^{(n+1)\log(1+\frac{1}{n})}$, the whole exponent of e is

$$\begin{aligned} & (n+1) \log \left(1 + \frac{1}{n}\right) - n - 1 + (a + b + \dots) \\ &= (n+1) \left\{ \frac{1}{n} - \frac{1}{2n^2} + \frac{1}{3n^3} - \dots \right\} - n - 1 + (a + b + \dots) \\ &= -n + 1 + \frac{1}{2} - \frac{1}{3n} + \frac{1}{4n^2} - \dots - n - 1 + (a + b + \dots) \\ &= -\frac{1}{2}(a + b + \dots). \end{aligned}$$

We have

$$\frac{1}{1} \cdot \frac{1}{3} + \frac{1}{2} \cdot \frac{1}{5} + \frac{1}{3} \cdot \frac{1}{7} + \text{const.} + \frac{1}{(2n+1)} + \text{terms in } \frac{1}{n}, \frac{1}{n^2}, \&c.$$

(the constant is in fact $= \frac{1}{2}\pi$, but the value is not required), hence $a = \text{const.} - \frac{1}{2n} + \text{terms in } \frac{1}{n^2}, \frac{1}{n^3}, \&c.$; as regards $b, c, \&c.$, there are no terms in $\frac{1}{n}$, but we have $b = \text{const.} + \text{terms in } \frac{1}{n^2}, \&c.$, $c = \text{const.} + \text{terms in } \frac{1}{n^2}, \&c.$ Hence the whole exponent of e is

$$= -n + C^* + \frac{1}{2} - \frac{1}{12n} + \text{terms in } \frac{1}{n^2}, \&c.$$

As in Mr Glaisher's investigation, it is shown that $e^{C^* + \frac{1}{2}} = \sqrt{(2\pi)}$, and hence neglecting the terms in $\frac{1}{n^2}, \&c.$, the final result is

$$\Pi n = \sqrt{(2\pi)} n^{n+\frac{1}{2}} e^{-n-\frac{1}{12n}}.$$

678.

ON A SYSTEM OF QUADRIC SURFACES.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 124, 125.]

The following theorem was communicated to me by Dr Klein; “given in regard to a quadric surface two sibi-reciprocal line-pairs, the two tractors (or lines meeting each of the four lines) form a sibi-reciprocal line-pair.” This may be presented under a more general form as a theorem relating to the tractors of any two line-pairs. In fact, if a given line-pair is taken to be sibi-reciprocal in regard to a quadric surface, we thereby establish only a four-fold relation between the coefficients of the surface, and the surface will still depend on five arbitrary parameters. Whence if two given line-pairs are taken to be each of them sibi-reciprocal in regard to one and the same quadric surface, we thereby establish only an eight-fold relation and the surface will still depend upon one arbitrary parameter. The theorem thus is: given any two line-pairs, then each of these, and also the pair of tractors, are sibi-reciprocal in regard to a singly infinite system of quadric surfaces.

The question arises, what is this system of quadric surfaces? It is, in fact, the system of surfaces having in common a skew quadrilateral constructed as follows: starting from the two given line-pairs, construct the two tractors, each of them intersected by the given line-pairs in two point-pairs; and on each tractor construct the double or sibi-reciprocal points of the involution thus determined; these double points are the vertices (those on the same tractor being opposite vertices) of the skew quadrilateral; which is consequently at once obtained by joining the two double points on the one tractor with the two double points on the other tractor. The construction is an immediate consequence of the following theorem: consider a skew quadrilateral, and drawing its two diagonals, take a pair of lines cutting each diagonal harmonically; these will be sibi-reciprocal in regard to any quadric surface through the skew quadrilateral.

The condition of passing through a skew quadrilateral is that of passing through a certain system of eight points; in fact, the eight points may be taken to be the four vertices and any four points on the four sides respectively. But observe that the system of the quadric surfaces through any eight points has the characteristics $(1, 2, 3)$; viz. there are in the system 1 surface passing through a given point, 2 touching a given line, 3 touching a given plane; the system of surfaces through the same skew quadrilateral has the characteristics $(1, 2, 1)$.

679.

ON THE REGULAR SOLIDS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 127–131.]

In a regular solid, or say in the spherical figure obtained by projecting such solid, by lines from the centre, on the surface of a concentric sphere, we naturally consider 1° the summits, 2° the centres of the faces, 3° the mid-points of the sides. But, imagining the five regular figures drawn in proper relation to each other on the same spherical surface, the only points which have thus to be considered are 12 points A , 20 points B , 30 points Θ , and 60 points Φ . These may be, in the first instance, described by reference to the dodecahedron; viz. the points A are the centres of the faces, the points B are the summits, the points Θ are the mid-points of the sides, and the points Φ are the mid-points of the diagonals of the faces (viz. there are thus 5 points Φ in each face of the dodecahedron, or in all 60 points Φ). But reciprocally we may describe them in reference to the icosahedron; viz. the points A are the summits, the points B the centres of the faces, the points Θ the mid-points of the sides, (viz. each point Θ is the common mid-point of a side of the dodecahedron and a side of the icosahedron, which sides there intersect at right angles), and the points Φ are points lying by 3's on the faces of the icosahedron, each point of the face being given as the intersection of a perpendicular $A\Theta$ of the face by a line BB , joining the centres of two adjacent faces and intersecting $A\Theta$ at right angles.

The points A lie opposite to each other in pairs in such wise that, taking any two opposite points as poles, the relative situation is as follows:

A	Longitudes
1	—
5	$0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ$
5	$36^\circ, 108^\circ, 180^\circ, 252^\circ, 324^\circ$
1	—

where the points A in the same horizontal line form a zone of points equidistant from the point taken as the North Pole. And the points B lie also opposite to

each other in such wise that, taking two opposite points as poles, the relative situation is as follows:

B	Longitudes
1	—
3	$0^\circ, 120^\circ, 240^\circ$
6	$(0^\circ, 120^\circ, 240^\circ) + 22^\circ 14'$
6	$(60^\circ, 180^\circ, 300^\circ) + 22^\circ 14'$
3	$60^\circ, 180^\circ, 300^\circ$
1	—

where the points B in the same horizontal line form a zone of points equidistant from the point taken as the North Pole. Neglecting the $3 + 3$ B points which lie adjacent to the poles, the remaining 14 points B may be arranged as follows ($\beta = 22^\circ 14'$ as above):

B	Longitudes
1	—
6	$120^\circ + \beta, 240^\circ + \beta; 120^\circ - \beta, 240^\circ - \beta$
6	$60^\circ + \beta, 180^\circ + \beta, 300^\circ + \beta; 60^\circ - \beta, 180^\circ - \beta, 300^\circ - \beta$
1	—

And taking the two poles separately with each system of the remaining poles, we have 2 systems each of 8 points B , which are, in fact, the summits of a cube (hexahedron); each point B taken as North Pole thus belongs to two cubes; but inasmuch as the cube has 8 summits, the number of the cubes thus obtained is $20 \times 2 \div 8, = 5$; viz. the 20 points B form the summits of 5 cubes, each point of course belonging to 2 cubes.

It is to be added that, considering the 5 points B which form a face of the dodecahedron, any diagonal BB of this dodecahedron is a side of a cube. We have thus $12 \times 5, = 60$, the number of the sides of the 5 cubes.

It is at once seen that the centres of the faces of a cube are points Θ , and that the mid-points of the sides of the cube are points Φ .

To each cube there corresponds of course an octahedron, the summits being points Θ , the centres of the faces points B , and the mid-points of the sides points Φ ; thus, for the five octahedra the summits are the $5 \times 6, = 30$, points Θ ; the centres of the faces are $5 \times 8, = 40$, points B (each point B being thus a centre of face for two octahedra), and the mid-points of the sides being the $5 \times 12, = 60$, points Φ .

Finally, considering the 8 points B which belong to a cube, we can, in four different ways, select thereout 4 points B which are the summits of a tetrahedron:

the remaining 4 points B are then the centres of the faces, and the mid-points of the sides are points Θ : there are thus $5 \times 4, = 20$, tetrahedra having 20×4 summits which are the 20 points B each 4 times; 20×4 centres of faces which are the 20 points B each 4 times; and 20×6 mid-points of sides which are the 30 points Θ each 4 times.

It thus appears that, as mentioned above, the five regular figures depend only on the points A, B, Θ , and Φ .

We might take as poles two opposite points A, B, Θ , or Φ ; and in each case determine in reference to these the positions of the other points; but for brevity I consider only the case in which we take as poles two opposite points A . We have the following table:

Poles two opposite points A .

	N. P. D.	Longitudes
A_1	0°	0°
$5A_1$	$63^\circ 26'$	$0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ$
$5A_2$	$116^\circ 34'$	$36^\circ, 108^\circ, 180^\circ, 252^\circ, 324^\circ$
A_2	180°	
$5B_1$	$37^\circ 22'$	$36^\circ, 108^\circ, 180^\circ, 252^\circ, 324^\circ$
$5B_2$	$79^\circ 12'$	$36^\circ, 108^\circ, 180^\circ, 252^\circ, 324^\circ$
$5B_3$	$100^\circ 48'$	$0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ$
$5B_4$	$142^\circ 38'$	$0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ$
$5\Theta_1$	$31^\circ 43'$	$0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ$
$5\Theta_2$	$58^\circ 17'$	$36^\circ, 108^\circ, 180^\circ, 252^\circ, 324^\circ$
$10\Theta_3$	90°	$(0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ) + 18^\circ$
$5\Theta_4$	$121^\circ 43'$	$0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ$
$5\Theta_5$	$148^\circ 17'$	$36^\circ, 108^\circ, 180^\circ, 252^\circ, 324^\circ$
$5\Phi_1$	$13^\circ 16'$	$36^\circ, 108^\circ, 180^\circ, 252^\circ, 324^\circ$
$10\Phi_2$	$52^\circ 52'$	$(0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ) + 9^\circ 44'$
$10\Phi_3$	$68^\circ 10'$	$(0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ) + 13^\circ 35'$
$10\Phi_4$	$76^\circ 42'$	$0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ$
$10\Phi_5$	$103^\circ 18'$	$0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ$
$5\Phi_6$	$111^\circ 50'$	$36^\circ, 108^\circ, 180^\circ, 252^\circ, 324^\circ$
$5\Phi_7$	$127^\circ 8'$	$(36^\circ, 108^\circ, 180^\circ, 252^\circ, 324^\circ) + 13^\circ 35'$
$5\Phi_8$	$166^\circ 44'$	$(36^\circ, 108^\circ, 180^\circ, 252^\circ, 324^\circ) + 9^\circ 44'$
		$108^\circ.$

I add for greater completeness the following results, some of which were used in the calculation of the foregoing table. Considering successively (1) the tetrahedral triangle, summits 3 points B , centre a point B ; (2) the hexahedral square, summits 4 points B , centre a point Θ ; (3) the octahedral triangle, summits 3 points Θ , centre a point B ; (4) the icosahedral triangle, summits 3 points A , centre a point B ; (5) the dodecahedral pentagon, summits 5 points A , centre a point B ; and (6), what may be called the small pentagon, summits 5 points Φ lying within a dodecahedral pentagon, and having therewith the common centre B ; we may in each case write s the side, r the radius or distance of the centre from a summit, p the perpendicular or distance of the centre from a side. And the values then are

	s	r	p
Tet. $\triangle$	109° 30'	70° 30'	54° 45'
Hex. square	90	54 45	45
Oct. $\triangle$	63 26	37 22	35 15
Icos. $\triangle$	41 50	37 22	20 55
Dod. pentagon	15 30	13 16	31 43
Small pentagon			10 48

680.

ON THE HESSIAN OF A QUARTIC SURFACE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 141–144.]

The surface considered is

$$U = k^2 w^2 \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) - (x^2 + y^2 + z^2)^2 = 0,$$

or say

$$U = k^2 w^2 P - Q^2 = 0,$$

viz. this may be considered as the central inverse of the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

The values of the second derived functions, or terms of the Hessian determinant

$$\begin{vmatrix} a, & h, & g, & l \\ h, & b, & f, & m \\ g, & f, & c, & n \\ l, & m, & n, & d \end{vmatrix},$$

are

$$\begin{array}{cccc} \frac{k^2}{a^2} w^2 - 2Q - 4x^2, & -4xy, & -4xz, & \frac{2k^2}{a^2} wx \\ -4xy, & \frac{k^2}{b^2} w^2 - 2Q - 4y^2, & -4yz, & \frac{2k^2}{b^2} wy \\ -4xz, & -4yz, & \frac{k^2}{c^2} w^2 - 2Q - 4z^2, & \frac{2k^2}{c^2} wz \\ \frac{2k^2}{a^2} wx, & \frac{2k^2}{b^2} wy, & \frac{2k^2}{c^2} wz, & 2k^2 P \end{array}$$

and we thence have

$$bc - f^2 = \frac{k^4}{b^2c^2}w^4(3Q + 4y^2 + 4z^2) - \frac{2k^2w^2}{b^2c^2}(3Q + 4y^2 + 4z^2)Q + 4Q^2 - 24Q^3,$$

whence, forming the analogous quantities $-ca - g^2$, &c., it is easy to obtain

$$\begin{aligned} & abc - af^2 - bg^2 - ch^2 + 2fgh \\ &= -\frac{k^6w^6}{a^2b^2c^2} \left\{ 4Q^3 + 2Q^2 \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right) + 4 \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) P \right\} \\ & \quad + \frac{2k^4w^4}{a^2b^2c^2} \left\{ 12Q \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right) + 4 \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) \right\} \\ & \quad - 24Q^3, \end{aligned}$$

which is to be multiplied by $d, = k^2P$. And

$$\begin{aligned} & -[l^2(bc - f^2) + m^2(ca - g^2) + n^2(ab - h^2) \\ & \quad + 2mn(gh - af) + 2nl(hf - bg) + 2lm(fg - ch)] \\ &= -\frac{4k^6w^6}{a^2b^2c^2} \left[Q \left\{ \frac{x^2}{a^2} \left(\frac{1}{b^2} + \frac{1}{c^2} \right) + \frac{y^2}{b^2} \left(\frac{1}{c^2} + \frac{1}{a^2} \right) + \frac{z^2}{c^2} \left(\frac{1}{a^2} + \frac{1}{b^2} \right) \right\} \right. \\ & \quad \left. + 4 \left(\frac{x^4}{a^2b^2c^2} \left(\frac{1}{b^2} + \frac{1}{c^2} \right) + \dots \right) - 4 \left(\frac{x^2y^2}{a^2b^2} \left(\frac{1}{a^2} + \frac{1}{b^2} \right) + \dots \right) \right] \\ & \quad + 4k^4w^4 \left[3Q \left(\frac{x^4}{a^4} + \frac{y^4}{b^4} + \frac{z^4}{c^4} \right) - 4P \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) \right. \\ & \quad \left. + 12 \left(\frac{x^4}{a^4}Q^{-1} + \dots \right) - 4P \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) Q + 8P^2Q \right. \\ & \quad \left. + 24P^2Q \right] \end{aligned}$$

which is

$$\begin{aligned} & 4k^6w^6 \Delta P \\ & + 8w^4 \left[-48Q + \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right) Q + 32P^2Q \right]. \end{aligned}$$

Hence, uniting the two parts, we have

$$\begin{aligned}
 H &= 8w^4 \left(-\frac{8}{a^2b^2c^2} P \right) \\
 &+ 2k^4w^4 \left\{ -6 \left(\frac{x^4}{a^4b^2c^2} + \frac{y^4}{a^2b^4c^2} + \frac{z^4}{a^2b^2c^4} \right) PQ \right. \\
 &\quad \left. - \frac{24Q^3}{a^2b^2c^2} + 12 \left(\frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4} \right) P \right\} \\
 &+ 2k^4w^4 \left\{ 12 \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right) PQ \right. \\
 &\quad \left. - 48 \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) Q^2 \right. \\
 &\quad \left. + 24P^2Q \right\} \\
 &+ k^2w^2(-24PQ^3).
 \end{aligned}$$

Writing herein $Q^2 = k^2w^2P - U$, and transposing all terms which contain U , we have

$$\begin{aligned}
 H &= k^6w^6P \left\{ -\frac{8}{a^2b^2c^2} P + 2 \left(\frac{1}{a^2b^2} + \frac{1}{b^2c^2} + \frac{1}{c^2a^2} \right) Q - 4 \left(\frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4} \right) \right\} \\
 &- 48k^4w^4P^2 + 12k^4w^4 \left[-\frac{2Q^2}{a^2b^2c^2} + 2 \left(\frac{1}{a^2b^2} + \frac{1}{b^2c^2} + \frac{1}{c^2a^2} \right) Q - 8k^2w^2 \left(\frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4} \right) P - 24k^2w^2Q \right]
 \end{aligned}$$

where, in the term in $\{ \}$, the last four lines are

$$\begin{aligned}
 &= -16 \left(\frac{1}{a^2b^2} + \frac{1}{b^2c^2} + \frac{1}{c^2a^2} \right) Q \\
 &\quad - 32 \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right).
 \end{aligned}$$

Hence, writing for shortness

$$\Theta = \dots - \frac{32}{a^2b^2c^2}w^2 + 2k^2w^2 \left(\frac{1}{a^2b^2} + \frac{1}{b^2c^2} + \frac{1}{c^2a^2} \right) Q - 4k^2w^2 \left(\frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4} \right) - 32PQ.$$

we have

$$H = 12k^6w^6P \cdot \Theta = 16k^4w^4 \left\{ -\frac{32}{a^2b^2c^2}w^2 + 2\left(\frac{1}{a^2b^2} + \frac{1}{b^2c^2} + \frac{1}{c^2a^2}\right)Q - 4\left(\frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4}\right) \right\}.$$

Hence, recollecting that $U = k^2w^2P - Q^2$, the Hessian curve of the order 32 breaks up into

$$U = 0, \quad w^4 = 0, \quad \text{that is, } Q^2 = 0, \quad w^4 = 0, \quad \text{or the nodal conic,}$$

$$w = 0, \quad Q = 0, \quad 8 \text{ times (order 16),}$$

$$U = 0, \quad P = 0, \quad \text{that is, } Q^2 = 0, \quad P = 0, \quad \text{or the quadriquadric,}$$

$$P = 0, \quad Q = 0, \quad 2 \text{ times (order 8),}$$

and into a curve (order 8) which is

$$-\frac{32}{a^2b^2c^2}w^2 + 2k^2w^2\left(\frac{1}{a^2b^2} + \frac{1}{b^2c^2} + \frac{1}{c^2a^2}\right)Q - 4k^2w^2\left(\frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4}\right) = 0,$$

viz. this, the intersection of the surface with a quadric surface, is the proper Hessian curve.

681.

ON THE DERIVATIVES OF THREE BINARY QUANTICS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 157–168.]

For a reason which will appear, instead of the ordinary factorial notation, I write $\{\alpha 012\}$ to denote the factorial $\alpha \cdot \alpha + 1 \cdot \alpha + 2$, and so in other cases; and I consider the series of equations

$$\begin{aligned} (1) &= X, \\ (2) &= (\{\alpha 0\}, \{\beta 0\})(Y, Y'), \\ (3) &= (\{\alpha 01\}, 2\{\alpha 1\}\{\beta 1\}, \{\beta 01\})(Z, Z', Z''), \\ (4) &= (\{\alpha 012\}, 3\{\alpha 12\}\{\beta 2\}, 3\{\alpha 2\}\{\beta 12\}, \{\beta 012\})(W, W', W'', W'''), \\ &\quad \&c. \end{aligned}$$

where

$$\begin{aligned} X &= Y + Y', \\ Y &= Z + Z', \quad Y' = Z' + Z'', \\ Z &= W + W', \quad Z' = W' + W'', \quad Z'' = W'' + W''', \\ &\quad \&c. \end{aligned}$$

We have thus a series of linear equations serving to determine $X; Y, Y'; Z, Z', Z''; W, W', W'', W''';$ &c. We require in particular the values of $X; Y, Y'; Z, Z''; W, W''';$ &c., and I write down the results as follow:

$$\begin{aligned} X &= (1), \\ \{\alpha + \beta\}Y &= \{\alpha 0\}(1), \quad +1, \\ \{\alpha + \beta\}Y' &= \{\alpha 0\}(1), \quad -1; \end{aligned}$$

$$\begin{aligned}
 &\{\alpha + \beta 2\}(1), \quad \{\alpha + \beta 1\}(2), \quad \{\alpha + \beta 01\}(3), \\
 &\{\alpha + \beta 012\}Z = \{\beta 01\}, \quad +2\{\beta 1\}, \quad +1, \\
 &\{\alpha + \beta 012\}Z'' = \{\alpha 01\}, \quad -2\{\alpha 1\}, \quad +1; \\
 &\{\alpha + \beta 34\}(1), \quad \{\alpha + \beta 4\}(2), \quad \{\alpha + \beta 03\}(3), \quad \{\alpha + \beta 01\}(4); \\
 &\{\alpha + \beta 01 \dots 4\}W = \{\beta 012\}, \quad +3\{\beta 12\}, \quad +3\{\beta 2\}, \quad +1, \\
 &\{\alpha + \beta 01 \dots 4\}W''' = \{\alpha 012\}, \quad -3\{\alpha 12\}, \quad +3\{\alpha 2\}, \quad -1; \\
 &\{\alpha + \beta 456\}(1), \quad \{\alpha + \beta 156\}(2), \quad \{\alpha + \beta 036\}(3), \quad \{\alpha + \beta 015\}(4), \quad \{\alpha + \beta 012\}(5); \\
 &\{\alpha + \beta 01 \dots 6\}U = \{\beta 0123\}, \quad +4\{\beta 123\}, \quad +6\{\beta 23\}, \quad +4\{\beta 3\}, \quad +1, \\
 &\{\alpha + \beta 01 \dots 6\}U'''' = \{\alpha 0123\}, \quad -4\{\alpha 123\}, \quad +6\{\alpha 23\}, \quad -4\{\alpha 3\}, \quad +1;
 \end{aligned}$$

&c.

read

$$\alpha + \beta \cdot Y = \beta(1) + (2),$$

$$\alpha + \beta \cdot Y' = \alpha(1) - (2),$$

$$\alpha + \beta \cdot \alpha + \beta + 1 \cdot \alpha + \beta + 2 \cdot Z = \beta \cdot \beta + 1 \cdot \alpha + \beta + 2 \cdot (1) + 2 \cdot \beta + 1 \cdot \alpha + \beta + 1 \cdot (2) + \alpha + \beta \cdot (3),$$

$$\alpha + \beta \cdot \alpha + \beta + 1 \cdot \alpha + \beta + 2 \cdot Z'' = \alpha \cdot \alpha + 1 \cdot \alpha + \beta + 2 \cdot (1) + 2 \cdot \alpha + 1 \cdot \alpha + \beta + 1 \cdot (2) + \alpha + \beta \cdot (3),$$

&c.,

the law being obvious, except as regards the numbers which in the top lines occur in connexion with $\alpha + \beta$ in the $\{ \}$ symbols. As regards these, we form them by successive subtractions as shown by the diagrams

$$\begin{array}{c|c|c}
 3 \ 4 & 4 \ 5 \ 6 & 5 \ 6 \ 7 \ 8 \\
 2 \ 1 \ 4 & 3 \ 1 \ 5 \ 6 & 4 \ 1 \ 6 \ 7 \ 8 \\
 1 \ 1 \ 0 \ 3 & 1 \ 2 \ 0 \ 3 \ 6 & 1 \ 3 \ 0 \ 3 \ 7 \ 8 \\
 & 2 \ 1 \ 0 \ 1 \ 5 & 2 \ 2 \ 0 \ 1 \ 5 \ 8 \\
 & 3 \ 0 \ 1 \ 2 & 3 \ 1 \ 0 \ 1 \ 2 \ 7 \\
 & & 4 \ 0 \ 1 \ 2 \ 3
 \end{array}$$

and the statement of the result is now complete.

In part verification, starting from the Y -formulas (which are obtained at once), assume

$$\{\alpha + \beta 2\}(1), \quad \{\alpha + \beta 1\}(2), \quad \{\alpha + \beta 0\}(3),$$

we must have

$$\{\alpha + \beta 012\}Z = \lambda, \quad \lambda', \quad \lambda'',$$

$$\{\alpha + \beta 012\}Z' = \mu, \quad \mu', \quad \mu'',$$

$$\{\alpha + \beta 012\}Z'' = \nu, \quad \nu', \quad \nu'',$$

$$\{\alpha + \beta 012\}Y = \{\alpha + \beta 012\}Z' = \{\alpha + \beta 12\}(\{\beta 0\}, +1),$$

that is,

$$\{\alpha + \beta 012\}Z' + Z'' = \{\alpha + \beta 12\}Y', \quad = \{\alpha + \beta 12\}(\{\alpha 0\}, -1),$$

$$\{\alpha + \beta 2\}X + X' = \{\alpha + \beta 12\}\{\beta 0\},$$

$$\{\alpha + \beta 2\}X' + X'' = \{\alpha + \beta 12\}\{\alpha 0\},$$

and further

$$\{\alpha + \beta 2\}(\{\alpha 01\}, -2\{\alpha 1\}\{\beta 1\}, \{\beta 01\})(X, X', X'') = 0,$$

or, what is the same thing,

$$X + X' = \{\alpha + \beta 1\}\{\beta 0\},$$

$$X' + X'' = \{\alpha + \beta 1\}\{\alpha 0\},$$

$$(\{\alpha 01\}, -2\{\alpha 1\}\{\beta 1\}, \{\beta 01\})(X, X', X'') = 0.$$

And in like manner we have

$$\mu + \mu' = \{\alpha + \beta 2\} \cdot 1,$$

$$\mu' + \mu'' = \{\alpha + \beta 2\} \cdot 1,$$

$$(\{\alpha 01\}, -2\{\alpha 1\}\{\beta 1\}, \{\beta 01\})(\mu, \mu', \mu'') = 0;$$

and

$$\nu + \nu' = 0, \quad \nu' + \nu'' = 0,$$

$$(\{\alpha 01\}, -2\{\alpha 1\}\{\beta 1\}, \{\beta 01\})(\nu, \nu', \nu'') = 0.$$

We hence find without difficulty

$$\lambda, \mu, \nu = \beta \cdot \beta + 1 \cdot \alpha + \beta + 2 \cdot \beta + 1, \quad +1, = \{\beta 01\}, \quad 2\{\beta 1\}, \quad 1,$$

$$\lambda', \mu', \nu' = \alpha \cdot \beta, \quad \alpha - \beta, \quad -1, = \{\alpha 0\}\{\beta 0\}, \quad \alpha - \beta, \quad -1,$$

$$\lambda'', \mu'', \nu'' = \alpha \cdot \alpha + 1, \quad -2 \cdot \alpha + 1, \quad 1, = \{\alpha 01\}, \quad -2\{\alpha 1\}, \quad 1;$$

viz. for verification of the λ -equations we have

$$\beta \cdot \beta + 1 \cdot \alpha + \beta, \quad \text{that is, } \alpha + \beta \cdot \beta = \{\alpha + \beta 1\}\{\beta 0\},$$

$$\alpha \cdot \beta + \alpha \cdot \alpha + 1, \quad \text{that is, } \alpha + 1 + \beta \cdot \alpha = \{\alpha + \beta 1\}\{\alpha 0\},$$

and

$$(\alpha \cdot \alpha + 1, \quad -2 \cdot \alpha + 1 \cdot \beta + 1, \quad \beta \cdot \beta + 1)(\beta \cdot \beta + 1, \quad \alpha \cdot \beta, \quad \alpha \cdot \alpha + 1) = 0,$$

that is,

$$\alpha \cdot \alpha + 1 \cdot \beta \cdot \beta + 1 \cdot -2 \cdot \alpha + 1 \cdot \beta + 1 \cdot \alpha \cdot \beta + \beta \cdot \beta + 1 \cdot \alpha \cdot \alpha + 1 = 0;$$

and similarly the μ - and ν -equations may be verified.

We have thus for the Z 's the equations

$$\{\alpha + \beta 2\}(1), \quad \{\alpha + \beta 1\}(2), \quad \{\alpha + \beta 0\}(3),$$

$$\{\alpha + \beta 012\}Z = \{\beta 01\}, \quad 2\{\beta 1\}, \quad +1,$$

$$\{\alpha + \beta 012\}Z' = \{\alpha 0\}\{\beta 0\}, \quad \alpha - \beta, \quad -1,$$

$$\{\alpha + \beta 012\}Z'' = \{\alpha 01\}, \quad -2\{\alpha 1\}, \quad +1,$$

which include the foregoing expressions for Z and Z'' .

We may then take the expressions for the W 's to be

$$\{\alpha + \beta 34\}(1), \quad \{\alpha + \beta 14\}(2), \quad \{\alpha + \beta 03\}(3), \quad \{\alpha + \beta 01\}(4),$$

$$\{\alpha + \beta 0123\}W = \lambda,$$

$$\{\alpha + \beta 0123\}W' = \lambda',$$

$$\{\alpha + \beta 0123\}W'' = \lambda'',$$

$$\{\alpha + \beta 0123\}W''' = \lambda''',$$